

Within-language lexical interference can be resolved in a similar way to between-language interference

Iva Ivanova^{*}, Dacia Carolina Hernandez

Department of Psychology, University of Texas at El Paso, USA

ARTICLE INFO

Keywords:

Language production
Bilingual language control
Picture naming
Mixing costs
Switching costs
Dominance effect

ABSTRACT

This study asks if monolinguals can resolve lexical interference within a language with mechanisms similar to those used by bilinguals to resolve interference across languages. These mechanisms are known as *bilingual language control*, are assumed to be at least in part top-down, and are typically studied with cued language mixing, a version of which we use here. Balanced (Experiment 1) and nonbalanced Spanish-English bilinguals (Experiment 2) named pictures in each of their languages. English monolinguals from two different American cities (Experiments 3 and 4) named pictures in English only with either basic-level (e.g., *shoe*) or subordinate names (e.g., *sneaker*). All experiments were identically structured and began with blocked naming in each language or name type, followed by trial-level switching between the two languages or name types, followed again by blocked naming. We analyzed switching, mixing and (introduced here) post-mixing costs, dominance effects and repetition benefits. In the bilingual experiments, we found some signs of dominant deprioritization, the behavioral hallmark of bilingual language control: larger costs for dominant- than for nondominant-language names. Crucially, in the monolingual experiments, we also found signs of dominant deprioritization: larger costs for basic-level than for subordinate names. Unexpectedly and only in the monolingual experiments, we also found a complete dominance reversal: Basic-level names (which otherwise behaved as dominant) were produced more slowly overall than subordinate names. Taken together, these results are hard to explain with the bottom-up mechanisms typically assumed for monolingual interference resolution. We thus conclude that top-down mechanisms might (sometimes) be involved in lexical interference resolution not only between languages but also within a language.

1. Introduction

It is near impossible to hear a Spanish-English bilingual say “mesa” when they intend to say “table”: Bilinguals possess efficient and well-attested language control mechanisms to avoid saying words in the wrong language when they may not be understood (Gollan, Sandoval, & Salmon, 2011; Poulish, 1999). But control of word selection seems no less necessary when conversing in a single language: Saying “dog” instead of “poodle” might end up ambiguous, “chair” instead of “table” puzzling, and “half-assed” instead of “inadequate” plain unwise in view of one’s chances of getting most jobs. Can monolinguals use within a language the same lexical interference resolution mechanisms that bilinguals use across languages? That is, are there language- and speaker-general mechanisms of control over word selection that can be applied both across and within languages? We address these questions here by tracing whether established behavioral signatures of language control in

bilinguals generalize to monolinguals naming pictures with basic-level (e.g., *shoe*) and subordinate names (e.g., *sneaker*) in their only language.

In asking these questions, we acknowledge that there are many differences between the language and cognitive systems of individuals who typically use two languages to communicate and those who typically use one. There is a lot of evidence that bilingualism brings cognitive and neural adaptations (Abutalebi & Green, 2007; Bialystok, Craik, & Freedman, 2007; Bialystok, Craik, & Luk, 2008; Bialystok & Martin, 2004; Costa, Hernández, & Sebastián-Gallés, 2008; but see Von Bastian, Souza, & Gade, 2016; Kousaie & Phillips, 2012; Paap & Greenberg, 2013). Further, two languages differ from each other in phonology and syntax (a difference absent within a single language), and they impose the need to “negotiate” which language is spoken when. The idea behind this study was not to underplay any of these differences, or the many benefits of being able to communicate in two languages. We (similarly to Dylman & Barry, 2018) simply aimed to see if there could be a certain

^{*} Corresponding author at: 500 W. University Ave., Department of Psychology, University of Texas at El Paso, El Paso, TX 79968, USA.

E-mail address: imivanova@utep.edu (I. Ivanova).

commonality of mechanism applied to lexical representations within and between languages. Even if that were the case, there would still be an important difference between monolinguals and bilinguals: Such a process would be applied a lot more frequently by bilinguals (to prevent both between- and within-language interference) than by monolinguals (who only need to prevent within-language interference).

For the sake of clarity, we talk about dominant and nondominant languages instead of first and second languages (or L1 and L2) because the dominant language is typically different from the first-learned language for the population we study here. For consistency, we also use these terms to refer to L1 and L2 as used by other authors (Christoffels, Ganushchak, & La Heij, 2016; Dylman & Barry, 2018 and Kroll & Stewart, 1994).

1.1. Resolving cross-language lexical interference in bilinguals

Preventing interference from the words belonging to the non-target language in bilingual speech is necessary under the assumptions that bilinguals' two languages have a common conceptual system (for a review, see Francis, 2005), and shared concepts activate both a target word and its non-target translation equivalent (Colomé, 2001; Costa, Caramazza, & Sebastian-Galles, 2000; Hoshino & Kroll, 2008). Most accounts assume that translation equivalents then compete for selection with target words (following e.g., Roelofs, 2003). This makes it non-trivial to explain how bilinguals manage to restrict production to a target language when required by the communicative situation (and produce extremely few wrong-language intrusions: Gollan et al., 2011; Poulisse, 1999).

Cross-language interference prevention in bilinguals is known as language control. Bilingual language control may function in several different ways. In the most established view, bilinguals avoid wrong-language intrusions by inhibiting the non-response language (Inhibitory Control Model: Green, 1998). A Supervisory Attentional System (as in Norman & Shallice, 1986) acts on *language task schemas* (the mental structure of different language activities such as "Speak Language X") that in turn send inhibition to nontarget-language schemas (global inhibition) and nontarget-language lexical representations (local inhibition). Inhibition is proportional to the strength of the language it acts on: A more dominant language (more likely to cause interference) is inhibited more strongly than a nondominant language (less likely to cause interference). In non-inhibitory accounts, language control happens through context-dependent adjustments of the activation levels (Finkbeiner, Gollan, & Caramazza, 2006; La Heij, 2005; Poulisse & Bongaerts, 1994) or lexical selection thresholds (Branzi, Martin, Abutalebi, & Costa, 2014) of words in either or both the target and non-target languages in situations of dual language activation (see also Grosjean, 2001; Philipp, Gade, & Koch, 2007; Runqvist, Strijkers, & Costa, 2019). Non-inhibitory language control can also be achieved if translation equivalents do not compete for selection (Costa, Sebastián-Gallés, Miozzo, & Caramazza, 1999), a state possibly attainable only with high proficiency in at least two languages (Costa, Santesteban, & Ivanova, 2006). Regardless of the precise mechanism, bilingual language control is usually assumed to involve a top-down component (Green, 1998; for neural evidence, see Abutalebi et al., 2008; Blanco-Elorrieta & Pykkänen, 2016; Seo, Stocco, & Prat, 2018; overview in Abutalebi & Green, 2007; Calabria, Costa, Green, & Abutalebi, 2018). One exception is the account proposed by Runqvist, Strijkers, Alario, and Costa (2012), which we discuss below.

1.2. Indicators of bilingual language control

The most consistent behavioral manifestation of bilingual language control is a *deprioritization of the dominant language* for nonbalanced bilinguals in a dual-language context, relative to when the dominant language is produced without prior (recent) activation of the nondominant language. Dominant deprioritization amounts to slower responses

in timed picture naming (Branzi et al., 2014; Costa & Santesteban, 2004; Macizo, Bajo, & Paolieri, 2012; Meuter & Allport, 1999; Philipp et al., 2007; Verhoef, Roelofs, & Chwilla, 2009), reduced lexical accessibility in verbal fluency tasks (Van Assche, Duyck, & Gollan, 2013) and increased number of intrusion errors in paragraph reading (Gollan & Goldrick, 2016; Gollan, Schotter, Gomez, Murillo, & Rayner, 2014). We use the term "deprioritization" in a theory-neutral sense that includes the possibility for a "boost" of the nondominant language when it is target instead of a "hit" to the dominant language when it is non-target (e.g., Philipp et al., 2007; Runqvist et al., 2019).

Some of the most frequent and robust demonstrations of dominant deprioritization come from cued picture naming in two languages. Such tasks have produced three relatively consistent patterns of dominant deprioritization, each of which is considered in the literature a behavioral "signature" of language control. The first is *switching costs*. They are measured in a mixed-language number- or picture-naming task (e.g., Meuter & Allport, 1999; Costa & Santesteban, 2004; review in Declerck & Philipp, 2015). One version of this task involves (predictably or unpredictably) switching languages every few trials. Naming a picture in a language different from that of the preceding trial (i.e., switching) results in a reaction time cost known as switching cost,¹ assumed to result from the time needed to disengage from the preceding task and recuperate the current target language from a previously applied control mechanism. In the majority of studies, switching costs were larger for the dominant language of nonbalanced bilinguals, i.e., showed dominant deprioritization (Costa & Santesteban, 2004; Jin, Zhang, & Li, 2014; Macizo et al., 2012; Meuter & Allport, 1999; Peeters, Runqvist, Bertrand, & Grainger, 2014; Philipp et al., 2007; Verhoef et al., 2009; review in Bobb & Wodniecka, 2013; although some studies have found symmetrical switching costs for two languages of unequal dominance, Christoffels, Firk, & Schiller, 2007; Verhoef, Roelofs, & Chwilla, 2010, or larger switching costs for the non-dominant language, Declerck, Stephan, Koch & Philipp, 2015). Note that symmetrical switching costs for the two languages of *balanced* bilinguals (Costa et al., 2006; Costa & Santesteban, 2004) are predicted because two similarly dominant languages would necessitate language control mechanisms of similar strength.

Adding single-language blocked naming for the two languages to the switching task described above also allows to measure *mixing costs*. Mixing costs are determined by subtracting naming latencies during blocked naming from naming latencies on non-switch trials in a mixed-language block (i.e., naming a picture or digit in a given language after having named the previous picture or digit in the same language). Mixing costs are thought to reflect the global costs of maintaining readiness to produce two different name types in mixed blocks, and are considered another indicator of language control processes. Typically, mixing costs also show an asymmetry for non-balanced bilinguals, such that there are larger mixing costs in the dominant than in the non-dominant language(s) (Christoffels et al., 2007; de Bruin, Roelofs, Dijkstra, & FitzPatrick, 2014; Gollan & Ferreira, 2009; Mosca & Clahsen, 2016; Peeters & Dijkstra, 2018). Of note, switching costs and mixing costs appear to be complementary, such that only one marker shows dominant deprioritization in tasks where both are measured (mixing costs showed dominant deprioritization: Christoffels et al., 2007; de Bruin et al., 2014; Gollan & Ferreira, 2009; Mosca & Clahsen, 2016; Peeters & Dijkstra, 2018; switching costs showed dominant deprioritization: Declerck, Philipp, & Koch, 2013; Wang, Kuhl, Chen, & Dong, 2009). In balanced bilinguals, mixing costs should be symmetrical for the same reason as switching costs: Languages of similar dominance would need the application of similar amounts of control.

A third indicator of control processes during cued mixed-language

¹ The current discussion of language control markers is framed around naming latencies because error rates are often at ceiling and produce less consistent patterns.

picture naming is the *language dominance effect*, which refers to the overall speed difference between the dominant and the non-dominant language. Typically and expectedly, dominant naming is faster when it precedes non-dominant naming (Christoffels et al., 2007; for reviews, see Hanulová, Davidson, & Indefrey, 2011; Runnqvist, Strijkers, Sadat, & Costa, 2011), because dominant-language words are more robustly represented and are assumed to have higher resting activation or lower selection thresholds. However, in dual-language situations, the overall naming speeds for the two languages may become similar (Calabria, Hernández, Branzi, & Costa, 2012; Declerck et al., 2013, Experiment 3; Filippi, Karaminis, & Thomas, 2014; Fink & Goldrick, 2015; Kleinman & Gollan, 2016), or the non-dominant language may even be produced more quickly overall (a *reversed dominance effect*: Costa & Santesteban, 2004; Costa et al., 2006; Christoffels et al., 2007; Gollan & Ferreira, 2009; Li & Gollan, 2018; Declerck, Kleinman, & Gollan, 2020; Peeters & Dijkstra, 2018; Verhoef et al., 2009, 2010; trend in Mosca & Clahsen, 2016). Equalized or reversed dominance has been interpreted as global deprioritization of the dominant language. We note that the three language control indicators discussed so far are not independent of one another, and the three may pattern together differently across studies because of differences in tasks (e.g., length, number of stimuli, number of repetitions, switch-no switch ratio, stimulus valence) and participants' language history.

Finally, we reasoned that a task with single-language blocked naming both before and after trial-level switching (see Christoffels et al., 2016; Gollan, Kleinman, & Wierenga, 2014; Prior & Gollan, 2011) would allow measuring another type of costs. We thus introduce *post-mixing costs*: blocked naming latencies after trial-level switching minus blocked naming latencies before trial-level switching. These costs could be another marker of dominant deprioritization that reflects sustained effects of language mixing independent of the need for task-set reconfiguration.² They also have the advantage of comparing the same types of trial, blocked naming.

1.3. Resolving within-language lexical interference in monolinguals

The problem of shared concepts highly activating two lexical items is not limited to bilinguals. It also occurs for synonyms (albeit to a lesser extent because not all words have close synonyms), and, at least partially, to words that share aspects of meaning such as register equivalents (e.g., *dog* – *pooch*) or words of different levels of specificity (e.g., *dog* – *Dalmatian*). However, the bottom-up mechanisms proposed for monolingual lexical interference resolution (reviewed below) are qualitatively different from the top-down mechanisms typically assumed for bilinguals (e.g., Abutalebi et al., 2008; Abutalebi & Green, 2007; Blanco-Elorrieta & Pykkänen, 2016; Calabria et al., 2018; Green, 1998; Seo et al., 2018).

Within-language (monolingual) lexical interference resolution is explained by two broad types of account, with or without competition for selection. On the traditional account, words within a language compete for selection (and greater competition results in slower and more effortful production: e.g., Starreveld & La Heij, 1995; Wheeldon & Monsell, 1994). One proposed solution to keeping in check competitors' activation levels is lateral inhibition (inhibition originating from a particular representation in proportion to its own current activation, and not from an external system, such as in the bilingual Inhibitory Control Model of Green, 1998). Activated lemmas (abstract word representations) send lateral inhibition to other activated lemmas to prevent

² With this comparison, Christoffels et al. (2016) actually found post-mixing benefits, likely because the trial-level switching part of their task involved 33 stimulus repetitions, and thus 1650 total trials. Still, their findings reflect clear signs of dominant deprioritization, because not only were the benefits larger for the nondominant language, but dominance completely reversed from initial to final blocked naming.

interference (Cutting & Ferreira, 1999; Howard, Nickels, Coltheart, & Cole-Virtue, 2006). On an alternative account, there is no lateral inhibition, and instead the system examines all activated words and selects the one whose activation exceeds the activation of all others by some critical amount (Levelt, Roelofs, & Meyer, 1999; Roelofs, 1992). A further recent hypothesis is that the activation differences between target and non-target words that would produce competition result from a flexible criterion, adjusted in response to factors such as task goals (Nozari & Hepner, 2019). In the way we understand this account, goals would be evaluated, and the criterion set, by top-down decision-making processes – but competition itself would be resolved by the language-internal bottom-up processes described above.

Without competition, words' ease and speed of selection may hinge on the speed with which target and non-target words reach the (bottleneck) articulation stage, and with which non-target words can be rejected (Response-exclusion Account, Mahon, Costa, Peterson, Vargas, & Caramazza, 2007). There is also no competition for selection in the incremental learning model of Oppenheim, Dell, and Schwartz (2010), in which a naming instance (e.g., *horse*) strengthens the conceptual-lexical connections of that representation, and weakens the connections of all related representations (e.g., *cow*) in proportion to their relatedness. Of relevance here, no-competition accounts also involve mechanisms that are qualitatively different from the ones proposed for resolving between-language interference (e.g., Abutalebi & Green, 2007; Green, 1998).

1.4. Speaker- and language-general mechanisms for interference resolution?

But is it possible that monolinguals and bilinguals use the same control mechanisms within- and across languages? The idea of shared features of the monolingual and bilingual lexical selection systems has been put forward by several authors, but the evidence is mixed. On the one hand, Dylman and Barry (2018) showed evidence for commonalities between bilingual and monolingual lexical selection. Three groups of bilinguals (Spanish-English, Swedish-English and English-French) and two groups of British English monolinguals named pictures while ignoring superimposed distractor words. For bilinguals, there were targets and distractors in both their first and dominant, and in their second and weaker language. For monolinguals, targets and distractors were either common names (e.g., *dog*) or alternative, less frequent coordinate synonyms (e.g., *hound*). There was facilitation from both translation-equivalent and synonym distractors, but more weakly represented names (nondominant-language names for bilinguals and alternative synonyms for monolinguals) benefitted more from it than more robustly represented names (dominant-language names for bilinguals, and common synonyms for monolinguals). While acknowledging that translation equivalents and synonyms differ in some respects, the authors concluded that they are represented similarly at some level. These findings suggest that there can be similarities between single and dual language systems.³

On the other hand, Melinger (2018) showed that distractor words belonging to a different dialect (or a different register) produced interference, in contrast to the (translation-)facilitation effects in Dylman and Barry (2018) and Costa et al. (1999). Melinger (2018) thus concluded that dialects are represented differently from languages and draw on different control mechanisms, specific to within-language control. The opposing within-language distractor effects (competition from category-coordinate different dialect distractors and no competition from synonyms) were reconciled by Melinger (2021), who replicated them in the same experimental paradigm. She proposed that the effects depend on

³ Dylman and Barry's (2018) proposed architecture includes a C-system which performs control functions, among others, and is thus compatible with results from language switching studies.

the number of currently activated strong lexical competitors, which create a trade-off between lexical competition (when there are many lexical competitors) and conceptual facilitation (when there is only one lexical competitor). Important here, this proposal situates both types of effect within the same system, and attributes the different polarity of effects to representational differences.

Other studies have also concluded that single- and dual-language control use (at least in part) similar mechanisms. Declerck, Grainger, Koch, & Philipp (2017; Experiment 3) compared the switching costs between language and within language (via switching between picture naming and category naming). The two types of switching costs were correlated and the costs in naming latencies did not differ between the two tasks (although the costs in error rates did). Further, Kirk, Kempe, Scott-Brown, Philipp, and Declerck (2018) showed similar switch-cost patterns for dialect switching as for language switching. In contrast to Melinger (2018), they concluded that monolingual dialect speakers recruit control mechanisms in similar ways as bilingual speakers.

Further evidence for shared mechanisms of control of single and dual-language lexical selection was provided by Finkbeiner, Almeida, Janssen, and Caramazza (2006), who found dominant deprioritization in a single-language context by monolinguals. In this study, monolinguals reading “faster” words (e.g., *dog*, *house*, *stone*) incurred larger switching costs than reading “slower” words (e.g., *puppy*, *cottage*, *pebble*) during switching between word reading and color naming (which the authors interpreted as evidence against bilingual inhibitory control).

Evidence for single- and dual-language control mechanisms that are in part similar and in part different was shown by Declerck, Ivanova, Grainger, and Duñabeitia (2020). These authors compared the trial-level costs incurred by bilinguals when switching between their two languages (French and English) and the costs incurred by these same bilinguals when switching between formal and informal words (e.g., *dog* versus *pooch*) in their dominant language (French). The switching costs were comparable, symmetrical in both cases, and correlated. However, increasing the cue-to-stimulus interval reduced the between-language switching costs (a reduction the authors attributed to the opportunity for phonological preparation of the switch to the other language) but left unaffected the within-language switching costs (where there is no opportunity to benefit from the preparation of a different phonological system).

On the other hand, neural differences between processing in a dual- and a single-language context were shown by Abutalebi et al. (2008). Switching between naming objects in two languages caused increased activation in the left caudate and anterior cingulate cortex, relative to switching between naming objects and their associated verbs in a single language by German-French bilinguals. However, the need to prevent interference between object names and verbs (which are not used in the same grammatical context and thus might never compete for selection in real-life language use) might not be as high as between object names in two languages (which mean the same thing and fit within the same grammatical context).⁴

An insightful investigation of the nature of the possibly shared mechanisms of between- and within-language control was conducted by Runnqvist et al. (2012). These authors tested models of bilingual language control – the Inhibitory Control Model (Green, 1998) and language-specific lexical selection (Costa et al., 1999) – with the cumulative semantic interference (CSI) paradigm. They found that naming latencies increased with the naming of each subsequent category member – a cumulative semantic interference effect that was equivalent when the response language alternated within a run and when it was kept constant. These results are inconsistent with either of the bilingual control models because they predict that the interference effect would

be reduced in half in alternating runs (because inhibiting or not attending to non-response language representations would make them non-competitors). Instead, the authors argued that their results are most parsimoniously explained with the bottom up mechanisms proposed for monolingual lexical selection by Howard et al. (2006) and Oppenheim et al., 2010 (see also Runnqvist et al., 2019, for relevant discussion).⁵ This proposal successfully accounts for target selection within and across languages and is appealing in its parsimony.

However, we note that this proposal does not, without additional assumptions, seem to explain how dominant deprioritization occurs. This is because the strength of bottom-up mechanisms such as activation and lateral inhibition is assumed to depend on a word’s own activation levels. Nondominant-language representations would have lower activation levels than dominant language ones. Consequently, they would not be very strong competitors (and neither could they win a competition solely based on activation levels). They would also not be able to spread stronger lateral inhibition than the one coming from dominant-language representations (such that could disproportionately disadvantage dominant-language words). Going back to our review above, the only bottom-up mechanism able to explain dominant deprioritization in certain situations is the Response-exclusion Account of Mahon et al. (2007) – the assumption that when dominant representations reach the articulatory buffer too fast in a situation of potential conflict, they are subjected to enhanced scrutiny, disproportionately slowing them down. Still, this account also appeared unable to explain the totality of results from the current study; we explain why in the General Discussion.

1.5. The present study

We ask if monolinguals can use the same mechanism to prevent within-language interference from highly activated semantically-related lexical items as bilinguals use to prevent between-language interference from highly activated translation equivalents. The existing evidence on the issue is mixed. Here, we analyze the language control indicators of monolinguals performing a within-language switching task. Of main interest is if monolinguals will show signs of dominant deprioritization, a robust consequence of language control in bilinguals. To have a bilingual benchmark in the current context, we also analyzed the language control indicators of bilinguals who performed a between-language switching task that was designed to be maximally comparable to the within-language task. Balanced (Experiment 1) and non-balanced bilinguals (Experiment 2) named colored photographs in English and Spanish, and monolinguals from two different American cities (Experiments 3 and 4) named photographs with basic-level (e.g., *shoe*) and subordinate names (e.g., *sneaker*) in English. The monolinguals in Experiment 4 were from the same city as the bilinguals. The between- and within-language tasks were identically structured and had a *sandwich* design. Each task began with blocked naming in each language or name type (*pre-mix blocks*), followed by a *mixed block* (trial-level switching between languages or name types), followed again by blocked naming in each language or name type (*post-mix blocks*).

For each experiment, the measure of interest is naming latencies, and we analyze the previously discussed indicators of language control: switching and mixing costs (to which we add an exploration of post-mixing costs), and the language or name-type dominance effect. In nonbalanced bilinguals, all of these markers index the presence of language control mechanisms through dominant deprioritization, and at least one of them should show dominant deprioritization when all three

⁴ Also, there was no overall difference between the three contexts in the behavioral data, and measures of deprioritization were not a focus in this study (and were not reported).

⁵ These mechanisms are lateral inhibition, whereby cross-language competition would be treated as an instance of semantic competition (Howard et al., 2006) or a lexical booster (a mechanism boosting activation of targets and competitors by the same constant until reaching a critical difference), which would ensure that the target is selected independent of language (Oppenheim et al., 2010).

Table 1

Language history characteristics of participants in all experiments.

	Balanced bilinguals, Exp. 1	Nonbalanced bilinguals, Exp. 2	Monolinguals (San Diego), Exp. 3	Monolinguals (El Paso), Exp. 4
Age of first exposure in years				
English	6.1 (3.6)	^a 3.1 (3.0)	0.15 (0.74)	1.0 (2.2)
Spanish	1.0 (1.3)	^a 2.9 (3.9)	not collected separately	
Other	14.4 (4.1), <i>N</i> = 30	13.3 (4.4), <i>N</i> = 23	^b 11.4 (5.4), <i>N</i> = 46	^b 11.5 (6.4), <i>N</i> = 15
% Daily use now				
English	62% (19%)	^a 82% (17%)	95% (20%)	95% (17%)
Spanish	42% (21%)	^a 23% (20%)	not collected separately	
Other	12% (9%), <i>N</i> = 7	5% (5%), <i>N</i> = 3	^{b,c} 8% (12%), <i>N</i> = 4	^b 2% (3%), <i>N</i> = 4
% Daily use as a child				
English	35% (23%)	^a 70% (27%)	95% (20%)	95% (20%)
Spanish	66% (22%)	^a 38% (29%)	not collected separately	
Other	none reported	10%, <i>N</i> = 1	^b 6% (4%), <i>N</i> = 3	^b 1% (3%) <i>N</i> = 4
Self-rated proficiency				
English	9.0 (1.0)	^a 9.7 (0.6)	10.0 (0.03)	9.8 (1.3)
Spanish	8.5 (1.4)	^a 5.2 (2.2)	3.2 (1.7), <i>N</i> = 38	2.2 (1.7), <i>N</i> = 37
Other	3.2 (2.1), <i>N</i> = 26	2.8 (2.0), <i>N</i> = 19	2.3 (2.1), <i>N</i> = 22	2.5 (2.8), <i>N</i> = 23
Productive vocabulary (MINT, of 68)				
English	57.8 (4.8)	62.1 (2.8)	Not collected	62.9 (3.1)
Spanish	55.2 (4.5)	29.6 (10.5)	Not collected	6.1 (7.7)

Note: Standard deviations are provided in parentheses.

^a *N* = 62 (the language history questionnaires of seven participants could not be uniquely identified).^b Includes Spanish.^c One participant reported speaking a language other than English 25% of the time.

are measured. For balanced bilinguals, on the other hand, dominant deprioritization effects should be absent (or, to the extent that bilinguals are rarely perfectly balanced, at least detectably weaker than for non-balanced bilinguals). Thus, based on established effects in prior literature, we predict significant and symmetrical switching and mixing costs for balanced bilinguals and asymmetrical switching or mixing costs for nonbalanced bilinguals.

The empirical question of main interest in this study is how control markers would pattern for monolinguals. Following the logic of Dylman and Barry (2018), we predict that monolinguals would show dominant deprioritization (of the more dominant basic-level names) if they employ similar control mechanisms to bilinguals in the current context.

We consider that basic-level names are dominant because they tend to be more frequent and familiar, and subordinate names are nondominant because they tend to be less frequent and familiar, but we also analyze *repetition benefits* of each name to obtain further evidence of name dominance. Repetition benefits and repetition priming are typically attributed to both short- and long-term strengthening of connections between representations in the language system (e.g., between semantic and lexical, or lexical and phonological representations; e.g., Barry, Hirsh, Johnston, & Williams, 2001). Of interest here, weaker representations – both within language and across languages – benefit more from such strengthening than stronger representations (low-frequency names: Griffin & Bock, 1998; nondominant language names: Kleinman & Gollan, 2018; Misra, Guo, Bobb, & Kroll, 2012). The larger effect for weaker representations may be due to more room for improvement for representations further away from lexical selection thresholds, or larger adjustments of system weights (as in error-driven learning models, e.g., Oppenheim et al., 2010) when a repetition amounts to a larger percentage of the total times a word is used. We also assumed that basic-level and subordinate names compete for selection (since they interfere with each other in picture-word tasks, Hantsch, Jescheniak, & Schriefers, 2005; Kuipers, La Heij, & Costa, 2006).

After conducting separate analyses of each experiment, we also conducted exploratory statistical comparisons between experiments (reported in brief in the main text and in full in Appendix C). We note, however, that our main predictions (presence or absence of dominant deprioritization for monolinguals) hinge on the analyses of the

individual experiments. The comparison of main interest was between monolinguals and nonbalanced (rather than balanced) bilinguals because basic-level and subordinate names within a language should involve a similar (although perhaps less pronounced) representational imbalance to that between a dominant and a nondominant language. Balanced bilinguals were an important part of the study design as a demonstration that the current paradigm does not produce the same dominant deprioritization effects across the board but instead is sensitive to the theoretically predicted grading of effects for non-balanced and balanced bilinguals.

The present study adds to the investigation of the generality of language control systems in the following ways. First, it is the only study known to us to focus on dominant deprioritization within and across languages, a phenomenon not straightforwardly explained by the bottom-up models of monolingual lexical selection. Second, it provides a systematic examination of dominant deprioritization by analyzing four different markers of it (differently from previous studies of this type such as Declerck et al., 2017; Declerck, Kleinman, & Gollan, 2020, and Kirk et al., 2018, who looked only at switching costs). Third, it looks at within-language switching between two different name types without the presence of another task (differently from some previous studies of this type which involved other tasks: category naming in Declerck et al., 2017, color naming in Finkbeiner, Almeida, et al., 2006). Fourth, it compares bilinguals with monolinguals (and not bilinguals within themselves) because bilinguals may have transferred their between-language control mechanisms to within-language control (unlike Declerck et al., 2017, and Declerck, Ivanova, et al., 2020, who compared single- and dual-language performance for bilinguals only).

2. Experiment 1: language mixing in balanced bilinguals

2.1. Method

2.1.1. Participants

Fifty-six balanced Spanish-English bilinguals, undergraduates at the University of Texas at El Paso (UTEP), participated for course credit or pay (\$10/h). The originally planned sample size, 48 participants per experiment, was based on the maximum in the range of sample sizes in

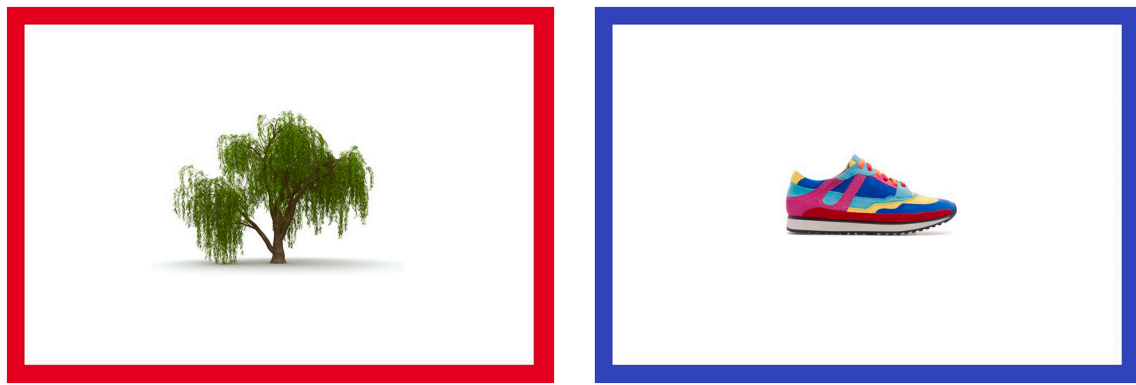


Fig. 1. Examples of experimental pictures in all experiments. Colored frames cued target name type. Left: For bilinguals, English (*tree*) or Spanish (*árbol*); for monolinguals, basic-level (*tree*) or subordinate (*willow*). Right: for bilinguals, Spanish (*zapato*) or English (*shoe*); for monolinguals, subordinate (*sneaker*) or basic-level (*shoe*).

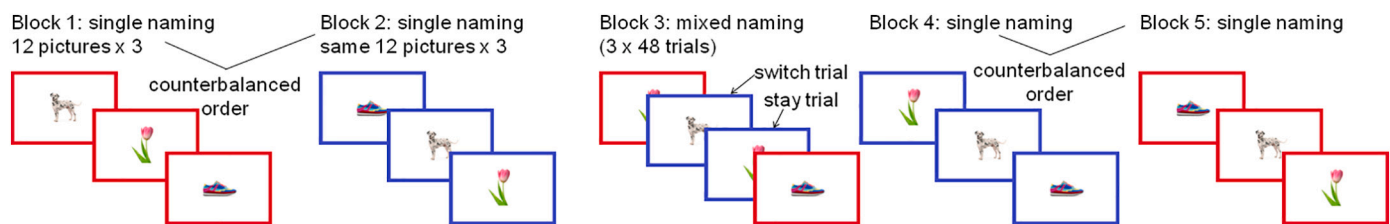


Fig. 2. Experimental paradigm. Color indicated language (English or Spanish) in Experiments 1 and 2, or name specificity (basic-level or subordinate) in Experiments 3 and 4.

published experiments of this type ($N = 12 - 48$; Costa & Santesteban, 2004; Declerck, Koch, & Philipp, 2012; Li & Gollan, 2018; Kirk et al., 2018; Peeters et al., 2014; Mosca & Clahsen, 2016; Philipp et al., 2007; Verhoeve et al., 2009).⁶ The final sample of 56 participants ($48 + 8$) was obtained after reinclusion of eight balanced bilinguals whose data was originally excluded due to an oversight.

Bilinguals' classification as balanced (Experiment 1) or nonbalanced (Experiment 2) was based on their English and Spanish scores on the Multilingual Naming Test (MINT), an objective productive vocabulary measure with 68 picture names of progressive difficulty (Gollan, Weissberger, Runnqvist, Montoya, & Cera, 2012). To be considered balanced, bilinguals had to be able to name not more than nine pictures in one language than in the other. On average, these bilinguals named three more pictures in English than in Spanish (range for English MINT scores minus Spanish MINT scores: $-9 - 9$). Participants' language history in all experiments is reported in Table 1.

Across Experiments 1 and 2 combined, 41 additional bilinguals were excluded from analyses. For most of them ($N = 29$), this was because they did not fit into the dominance group of either experiment. Twenty-two bilinguals were moderately English-dominant, naming on average 12 more pictures in English than in Spanish (range 10–15; the criterion for inclusion in Experiment 2 was naming more than 15 pictures in English than in Spanish). Part or all of the naming latency data for two of

these participants were lost. Seven bilinguals were Spanish-dominant, naming on average 14 fewer pictures in English than in Spanish (range $-10 - -23$). In addition, twelve bilinguals were excluded because of missing latency data for the whole ($N = 4$) or more than half of the experiment ($N = 5$), missing MINT scores ($N = 1$), or because it was impossible to uniquely match their latency data, error rates and MINT ($N = 2$).

2.1.2. Materials

The materials consisted of 12 colored photographs of common objects or animals inside a red or a blue frame to cue the target language (see Fig. 1 for examples). The English names of the photographs (*bag, bird, building, cheese, dress, fish, glass, horse, knife, painting, shoe, tree*) and their Spanish translations (*bolsa, pájaro, edificio, queso, vestido, pez, vaso, caballo, cuchillo, cuadro, zapato, árbol*) were all non-cognates and of medium to high frequency (English: $M = 64.48$, $SD = 26.17$; Spanish: $M = 47.62$, $SD = 26.90$, $t = 3.54$, $p = .005$). Frequency-per-million values were obtained from the movie-subtitles corpora SUBTLEX-US for American English (Brysbaert & New, 2009; <http://expsy.ugent.be/subtlexus/>), and SUBTLEX-ESP for Spanish (Cuetos, Glez-Nosti, Barbon, & Brysbaert, 2011; <http://crr.ugent.be/archives/679>). (Note that the SUBTLEX-ESP database largely reflects Castilian Spanish use and is thus only our best approximation for the Mexican-Spanish-speaking population tested here.) The English names were shorter than the Spanish names, reflecting language-wide differences in word length (English: $M = 4.08$, $SD = 1.73$; Spanish: $M = 5.50$, $SD = 1.38$; length in phonemes, $t = 4.71$, $p = .001$). The higher frequency and shorter length of the English versus the Spanish names (a limitation of these stimuli) were expected to enlarge the inherent proficiency differences between the two languages (for nonbalanced bilinguals). However, we assume that lexical characteristics and proficiency have a common origin and influence lexical selection in the same way (as proposed by the Weaker Links account, Gollan, Montoya, Fennema-Notestine, & Morris, 2005; see also Strijkers, Costa, & Thierry, 2010, about a common origin of frequency and cognate effects). That is, we assume that the higher frequency and

⁶ A post-hoc power analysis with G*Power (Faul et al., 2009) showed that 24 participants are necessary to detect an effect of $f = 0.25$ with 80% power in a repeated-measures ANOVA with one group and four measurements. (The effect sizes for the interactions reflecting the switch cost asymmetry for the non-balanced bilingual groups in Experiment 1 in Costa and Santesteban (2004) were $f = 0.78$ and $f = 1.13$. We are grateful to Mikel Santesteban for sharing the data.) Note that G*Power does not provide estimates for LMER analyses, but estimates for ANOVA analyses are conservative with enough observations per condition (at least 1600 are recommended to detect small effects in linear mixed-effects models, Brysbaert & Stevens, 2018); each of our experiments had at least 1728 observations.

Table 2

Error rates in all experiments.

Experiment	Block type						
Experiment 1	Single pre-mix	Mixed Stay	Mixing cost	Switch	Switching cost	Single post-mix	Post-mixing cost
English	2.04	2.72	0.69	3.49	0.77	0.65	-1.39
Spanish	3.13	2.71	-0.41	4.94	2.22	0.70	-2.43
Difference	1.09	-0.01		1.45		0.05	
Experiment 2							
English (dom.)	2.50	2.94	0.44	3.60	0.66	1.17	-1.33
Spanish (nondom.)	8.29	4.82	-3.47	6.29	1.47	2.98	-5.31
Difference	5.80	1.88		2.69		1.81	
Experiment 3							
Basic-level (dom.)	2.49	3.35	0.86	4.29	0.94	1.74	-0.75
Subordinate (nondom.)	1.85	2.05	0.20	4.05	1.99	0.75	-1.10
Difference	-0.64	-1.29		-0.24		-0.98	
Experiment 4							
Basic-level (dom.)	1.45	2.93	1.49	3.27	0.34	1.85	0.41
Subordinate (nondom.)	3.30	2.00	-1.30	3.51	1.52	1.45	-1.85
Difference	1.85	-0.94		0.24		-0.41	

shorter length of the English relative to the Spanish names added quantitative but not qualitative processing differences between the names in the two languages.

The number of stimuli was kept the same as that in the monolingual experiments (Experiments 3–4), which were more difficult to find. In any case, the number of stimuli used here is comparable to that used in the classical studies (ten numbers in Meuter & Allport, 1999; ten pictures in Costa & Santesteban, 2004, and Costa et al., 2006). The stimuli for the monolingual task were chosen first and replacements in the bilingual task were made to avoid cognates and low name agreement.

2.1.3. Design and procedure

The pictures were presented in five blocks in a sandwich design (Fig. 2) modeled after the design in Prior and Gollan (2011). At the beginning of each of the first three blocks, bilinguals read brief instructions in English about their task in the respective block, followed by practice consisting of naming all experimental pictures (once in each language in the mixed blocks). During practice, participants were informed of the intended name if they spontaneously produced a different one. The required language on each trial in the mixed blocks was indicated by the picture frame color (blue or red, counterbalanced across participants); the frames were also kept in the single-naming blocks.

In the first two *pre-mix* single-naming blocks (either only English or only Spanish), bilinguals named pictures on 12 practice and 36 (12 pictures × 3 repetitions) experimental trials per block. Language order was counterbalanced across blocks. A picture was never repeated on successive trials.

In the following three *mixed blocks*, bilinguals completed a single set of 24 practice trials followed by three sets of 48 trials of unpredictable switching between English and Spanish. Each of the three mixed blocks began with an additional dummy trial (excluded from analyses). Switch trials were half of all trials, and there were no more than four successive trials of the same type.

Language order in the two final *post-mix* single-naming blocks (36 trials each) was reversed from the first two single-naming blocks (A-B-mixed-B-A or B-A-mixed-A-B). The counterbalancing of language order in the pre-mix single-naming blocks (English first or Spanish first), item list order in the pre- and post-mix single-naming blocks (list order 1, 2 (pre-mix) + 3, 4 (post-mix) or list order 2, 1 (pre-mix) + 4, 3 (post-mix)), and frame color (blue or red) produced eight versions of the experiment, administered to a roughly equal number of participants (between 6 and 9). Twenty-nine bilinguals began with naming in Spanish, and 27 began with naming in English.

A trial had the following structure: A blank screen displayed for 700 ms was followed by an asterisk displayed for 700 ms, a blank screen

Table 3

LMER analyses of the data of Experiment 1.

Model	Predictors	Estimate	SE	t or z	p
Switching costs					
	Switching trial type (stay vs. switch)	46.00	8.40	5.48	< .001
	Language	16.45	16.70	.99	.34
	Switching trial type x Language	-15.34	10.04	-1.53	.13
Mixing costs					
	Mixing trial type (pre-mix vs. mixed-stay)	198.10	14.57	13.59	< .001
	Language	33.70	17.72	1.90	.07
	Mixing trial type x Language	-18.14	18.37	-.98	.33
Post-mixing costs					
	Single naming block type (pre-mix vs. post-mix)	66.94	12.56	5.33	< .001
	Language	29.06	15.78	1.84	.08
	Single naming block type x Language	-27.82	14.95	-1.86	.07
Repetition benefits					
	Block type (single vs. mixed)	-206.84	13.20	-15.67	< .001
	Language	21.30	15.18	1.40	.17
	Repetition (single: 1-3; mixed: 1-6)	-14.85	2.72	-5.46	< .001
	Block type x Language	2.90	10.41	.28	.78
	Block type x Repetition	-15.91	5.11	-3.11	.009
	Language x Repetition	-5.43	3.48	-1.56	.14
	Block type x Language x Repetition	-6.36	7.20	-.88	.39

Note: Dark grey shading indicates significant effects, light grey shading indicates marginal effects.

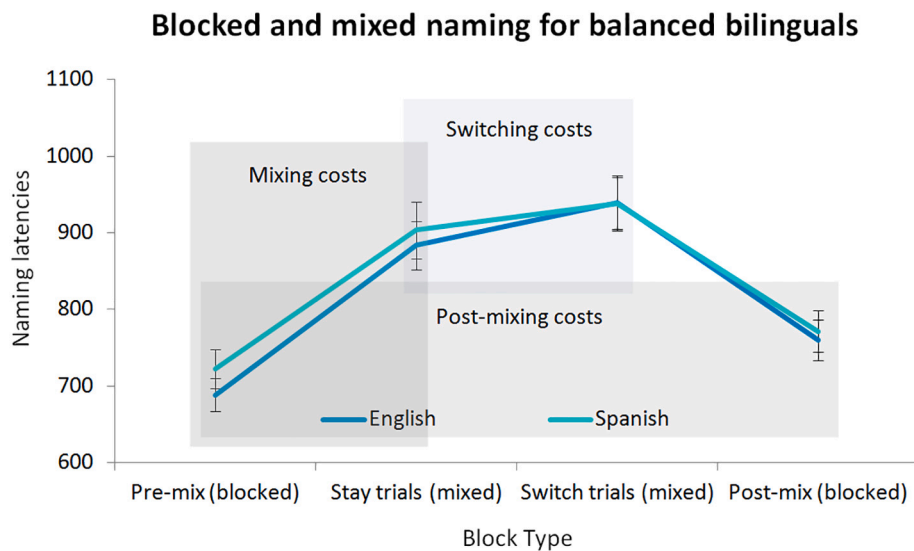


Fig. 3. By-participant mean naming latencies for each block type and language for balanced bilinguals in Experiment 1. Error bars represent 95% confidence intervals.

displayed for 500 ms, and the target picture displayed for 3000 ms or until the detection of a vocal response. A trained experimenter coded different-from-intended responses and voice-key inaccuracies in real time. The DMDX software for experimental presentation (Forster & Forster, 2003) recorded participants' vocal responses for subsequent verification.

The study procedures conformed to Federal guidelines for the protection of human subjects and were approved by the UTEP Institutional Review Board. All participants gave informed consent to participate prior to testing.

2.1.4. Coding and data analysis

Analyses excluded voice key inaccuracies and experimenter errors (826 trials, or 5.1% of the data) and, subsequently, outliers – response times that were slower than 3 standard deviations above the mean or faster than 300 ms (250 trials, or 1.6% of all data). Response times on correct trials and errors (which included different words in the same language, translation equivalents, disfluencies and failures to respond; together 405 trials or 2.5% of the data) were analyzed with linear or

logistic mixed effects regression modeling, respectively (LMER; Baayen, 2008; Jaeger, 2008). Error rates in all experiments are reported in Table 2, and analyses of error rates are reported in Appendix A.

Switching costs were assessed with a model comparing stay and switch trials in mixed blocks. The fixed predictors in this model were Switching trial type (stay trials, coded as -0.5 , switch trials, coded as 0.5 ; all other trials were coded as NA), Language (English, coded as -0.5 , Spanish, coded as 0.5) and their interaction. *Mixing costs* were assessed with a model comparing pre-mix single-naming trials and stay trials in mixed blocks. The fixed predictors in this model were Mixing trial type (pre-mix trials, coded as -0.5 , stay trials, coded as 0.5 ; all other trials were coded as NA), Language (coded as above) and their interaction. *Post-mixing costs* were assessed with a model comparing the pre-mix and post-mix single-naming blocks. The fixed predictors in this model were Single naming block type (pre-mix block, coded as -0.5 , post-mix block, coded as 0.5 ; the mixed block was coded as NA), Language (coded as above) and their interaction. *Repetition benefits* for each picture name in each language were assessed in a model with the fixed predictors Block type (single, coded as 0.5 , and mixed, coded as -0.5),

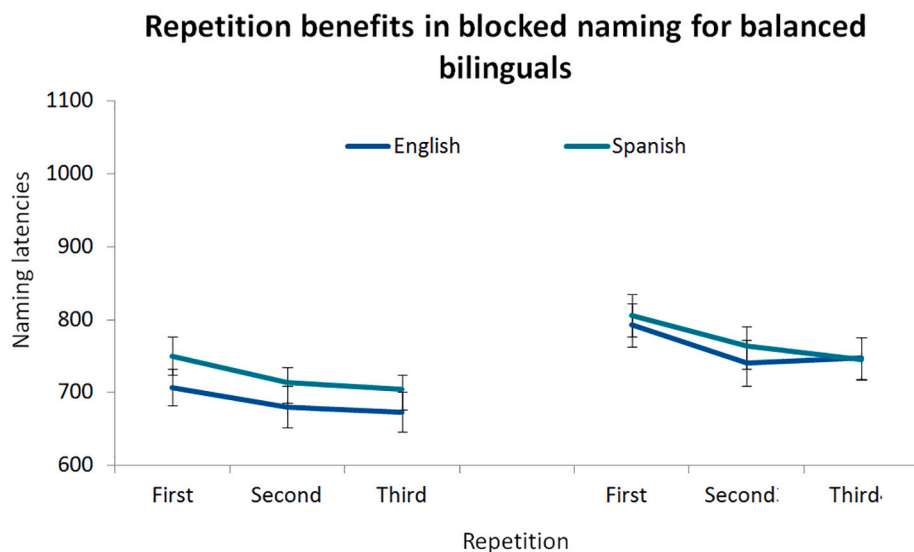


Fig. 4. By-participant mean naming latencies for each repetition in initial and final blocked naming for balanced bilinguals in Experiment 1. Error bars represent 95% confidence intervals.

Language (coded as above) and Repetition (a continuous mean-centered predictor). *Language dominance effects* were assessed in all models with the Language predictor. Additional supplementary analyses, where pertinent, are described in the Results section.

All models were implemented with the maximal random-effects structure justified by the design (Barr, Levy, Scheepers, & Tily, 2013) using the *lmer* or *glmer* functions in the *lmerTest* package (version 3.1–0, lme4 version 1.1–21) in R (version 3.6.1), using the Satterthwaite method to approximate denominator degrees of freedom for models with a continuous dependent variable. If the full random-effects model did not converge, the model was simplified by first removing random-effects correlations, and then removing step-wise the random effects accounting for least variance, with the restriction that random slopes were removed before random intercepts. The *bobyqa* optimizer was used to facilitate convergence.

Trial-level data for all experiments are publicly available at <https://osf.io/67yad/>.

2.2. Results

The results of all statistical models are reported in Table 3. By-participant mean naming latencies are plotted in Fig. 3 (per block type and language) and Fig. 4 (per repetition in initial and final blocked naming). Switching, mixing and post-mixing costs in all experiments are plotted in Fig. 11. The advantage gained by nondominant (Spanish or subordinate) names over dominant (English or basic-level) names from initial to final blocked naming in all experiments is plotted on Fig. 12.

2.2.1. Switching costs

There were significant switching costs: Response times were slower on switch than on stay trials in mixed blocks (Switching trial type was a significant predictor). In other words, balanced bilinguals named pictures more slowly when the immediately preceding picture was in a different language than when the immediately preceding picture was in the same language. As predicted, these costs did not differ across balanced bilinguals' languages (the interaction between Switching trial type and Language was not significant).

2.2.2. Mixing costs

The pattern for mixing costs was the same as for switching costs: There were significant mixing costs such that response times were slower on stay trials in mixed blocks than in pre-mix blocks (Mixing trial type was a significant predictor). In other words, balanced bilinguals named pictures in the same language as immediately preceding pictures more slowly within a block with frequent switches than within an initial block with no switches. As predicted, the mixing costs were equivalent for balanced bilinguals' two languages, i.e., symmetrical (the interaction between Mixing trial type and Language was not a significant predictor).

2.2.3. Post-mixing costs

Blocked naming after trial-level language switching also incurred a cost: Naming latencies in the post-mix single-naming blocks were slower than those in the pre-mix single-naming blocks (Single naming block type was a significant predictor). There was a trend for larger post-mixing costs for English than for Spanish (the interaction between Single naming block type and Language was marginally significant).

2.2.4. Repetition benefits

As expected, naming latencies got faster with repetition (Repetition was a significant predictor). This improvement was larger in single-naming blocks than in mixed blocks (the interaction between Block type and repetition was significant). To shed light on this interaction, we ran additional models on the separate data of the pre-mix, mixed and post-mix blocks with Language, Repetition and their interaction as fixed predictors. These analyses showed significant repetition benefits across all blocks (pre-mix block [Repetition predictor: *Estimate* = -20.90 , *SE* =

4.80 , $t = -4.36$, $p < .001$]; mixed block [Repetition predictor: *Estimate* = -6.99 , *SE* = 2.34 , $t = -2.99$, $p = .007$]; post-mix block [Repetition predictor: *Estimate* = -25.03 , *SE* = 8.63 , $t = -2.90$, $p = .01$]). Further, the repetition benefits were larger in Spanish than in English in the pre-mix block [Repetition x Language interaction: *Estimate* = -12.06 , *SE* = 5.65 , $t = -2.13$, $p = .04$], but not in the post-mix or mixed blocks [for the Repetition x Language interaction, both $ps > 0.6$].

2.2.5. Language dominance effects

Balanced bilinguals named pictures with a roughly similar speed in English and Spanish: The Language predictor was marginal in the mixing and post-mixing costs analyses (a trend for faster naming in English), and non-significant in the switching costs and repetition benefits analyses. Separate models on the data of the pre-mix and post-mix blocks aimed to see how the trend for faster naming in English developed in the course of the experiment. These models (with Language as the only fixed predictor) showed that the language difference was significant in the pre-mix blocks [*Estimate* = 43.22 , *SE* = 19.50 , $t = 2.22$, $p = .04$] but was not significant in the post-mix blocks [*Estimate* = 15.12 , *SE* = 15.15 , $t = 1.00$, $p = .33$]. In other words, there was an advantage of English over Spanish in the beginning of the experiment, but it disappeared by the end of the experiment. The marginally larger post-mixing costs for English (72 ms.) than for Spanish (49 ms.) indicated that this was at the expense of English (see Fig. 12).

2.3. Discussion

Balanced bilinguals in Experiment 1 showed significant switching and mixing costs, which, as predicted, did not differ across their two languages. Assuming that language control mechanisms act to allow less effortful production in the nondominant language, it is expected that similar levels of control would be applied to two languages of similar dominance because neither of their languages is much more likely to interfere than the other.

Some features of the data, however, indicated that the two languages of these bilinguals were not represented completely equivalently. There were larger repetition benefits for Spanish at the beginning of the experiment as well as a reduced difference between the languages at the end of the experiment relative to the beginning, at the expense of English. In view of the accepted interpretations of such effects in prior literature, both of these results suggest a slight dominance of English. This is in accordance with these bilinguals' language history reports, in which English (with an average self-rated proficiency of 9) was rated as slightly more dominant than Spanish (with an average self-rated proficiency of 8.5 [$t(55) = 2.34$, $p = .02$]). However, dominant deprioritization effects did not extend to a reversed dominance effect (unlike in Costa et al., 2006; Declerck, Kleinman, & Gollan, 2020; Kleinman & Gollan, 2016; Li & Gollan, 2018): Spanish was produced more slowly in initial blocked naming and with a similar speed to English in the subsequent blocks. This might be because our Spanish words were longer and of lower frequency than the English words.

Repetition benefits were asymmetrical at the beginning of the experiment but equivalent thereafter. Our interpretation of this finding is that the baseline accessibility of the names in the two languages was slightly different but it became more similar during their repeated retrieval in a mixed context. This pattern is consistent with the names' selection thresholds being adjusted in a mixed context (through inhibiting the more dominant one more strongly or through adjusting one or both thresholds directly).

3. Experiment 2: Language mixing in nonbalanced bilinguals

3.1. Method

3.1.1. Participants

Sixty-nine Spanish-English bilingual UTEP undergraduates dominant

Table 4
LMER analyses of the data of Experiment 2.

Model	Predictors	Estimate	SE	t or z	p
Switching costs					
	Switching trial type (stay vs. switch)	25.79	6.67	3.87	.003
	Language	36.11	15.74	2.29	.03
	Switching trial type x Language	13.26	11.12	1.19	.25
Mixing costs					
	Mixing trial type (pre-mix vs. mixed-stay)	172.44	13.41	12.86	< .001
	Language	74.83	13.83	5.41	< .001
	Mixing trial type x Language	-91.31	19.23	-4.75	< .001
Post-mixing costs on single naming					
	Single naming block type (pre-mix vs. post-mix)	50.01	13.99	3.58	< .001
	Language	87.77	12.65	6.94	< .001
	Single naming block type x Language	-66.56	13.17	-5.05	< .001
Repetition benefits					
	Block type (single vs. mixed)	-177.34	14.45	-12.28	< .001
	Language	56.01	12.34	4.54	< .001
	Repetition (single: 1-3; mixed: 1-6)	-12.26	3.36	-3.65	.002
	Block type x Language	45.15	14.67	3.08	.004
	Block type x Repetition	-15.93	6.67	-2.39	.03
	Language x Repetition	-3.43	3.49	-.98	.34
	Block type x Language x Repetition	-16.91	5.87	-2.88	.01

Note: Dark grey shading indicates significant effects, light grey shading indicates marginal effects.

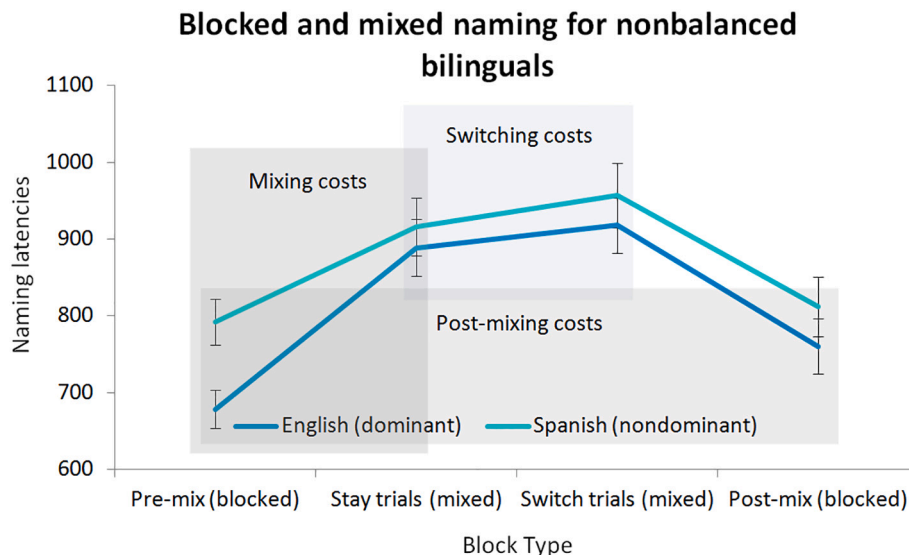


Fig. 5. By-participant mean naming latencies for each block type and language for nonbalanced bilinguals in Experiment 2. Error bars represent 95% confidence intervals.

in English participated for course credit or pay (\$10/h). To be classified as nonbalanced and dominant in English, bilinguals had to name at least 16 more pictures in English than in Spanish on the Multilingual Naming Test. On average, these bilinguals named 32 more pictures in English than in Spanish (range: 16–61). The final sample of 69 participants (48 + 21) was obtained after reinclusion of twenty-one balanced bilinguals whose data was originally excluded due to an oversight. Participants' characteristics are summarized in Table 1.

3.1.2. Materials, design, procedure and coding

These were the same as in Experiment 1. Experimental versions were administered to a similar number of participants (7–11). Thirty-one bilinguals began with naming in Spanish, and 38 began with naming in English.

3.1.3. Data analysis

The statistical models were the same as in Experiment 1. Analyses of naming latencies excluded 1079 responses (5.4%) as voice-key

inaccuracies, a further 414 responses (2.1%) as outliers and a further 812 responses (4.1%) as errors. Error rates are reported in Table 2. One participant completed the Spanish pre-mix block twice, and here we report this participant's performance the first time around.

3.2. Results

The results of the statistical models are reported in Table 4. By-participant mean naming latencies are plotted in Fig. 5 (per block type and language) and Fig. 6 (per repetition in initial and final blocked naming).

3.2.1. Switching costs

Switching costs for nonbalanced bilinguals in Experiment 2 patterned similarly to those of balanced bilinguals in Experiment 1. There were overall switching costs (Switching trial type was a significant predictor) and they were of similar magnitude for the two languages (the interaction between Switching trial type and Language was not

Repetition benefits in blocked naming for nonbalanced bilinguals

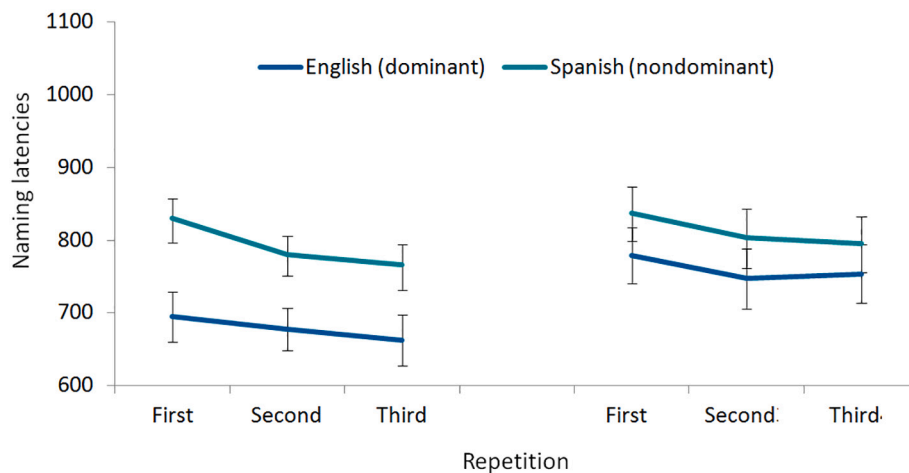


Fig. 6. By-participant mean naming latencies for each repetition in initial and final blocked naming for nonbalanced bilinguals in Experiment 2. Error bars represent 95% confidence intervals.

significant).

3.2.2. Mixing costs

Nonbalanced bilinguals in Experiment 2 showed significant mixing costs (Mixing trial type was a significant predictor). These costs were larger for English, the dominant language, than for Spanish, the nondominant language (the interaction between Mixing trial type and Language was significant).

3.2.3. Post-mixing costs

Single naming for nonbalanced bilinguals, as for balanced bilinguals, was slower after language mixing than before it (Single naming block type was a significant predictor). However, for nonbalanced bilinguals in Experiment 2 this cost was clearly asymmetrical, evidencing robust dominant deprioritization: It was larger for English, the dominant language, than for Spanish, the nondominant language (the interaction between Single naming block type and Language was significant).

3.2.4. Repetition benefits

As in Experiment 1, naming latencies sped up with repetition (Repetition was a significant predictor), and more so in the single-naming blocks than in the mixed block (the interaction between Block type and repetition was significant). Further, here repetition benefits were clearly asymmetrical between block and language (there was a three-way interaction between Repetition, Block type and Language). We then ran additional models on the separate data of pre-mix, mixed and post-mix blocks with Language, Repetition and their interaction as fixed predictors to shed light on this pattern. These analyses showed that repetition benefits were significant in the pre-mix block [Repetition effect: $Estimate = -23.49$, $SE = 6.22$, $t = -3.78$, $p = .003$], marginal in the post-mix block [$Estimate = -10.69$, $SE = 7.81$, $t = -1.37$, $p = .18$] and non-significant in the mixed block [$p > .18$]. The repetition benefits did not differ statistically between the two languages in any of the blocks [Repetition x Language interaction: all $ps > 0.11$], although numerically latencies in the single blocks sped up from the first to the third repetition twice as much in Spanish than in English (pre-mix block, Spanish: 64 ms; English: 32 ms; post-mix block, Spanish: 42 ms; English: 25 ms; Fig. 6).

3.2.5. Language dominance effects

Evident across all analyses, nonbalanced bilinguals named pictures faster in their dominant than in their nondominant language. Additional

models on the data of the pre-mix and post-mix blocks separately, with Language as a fixed predictor, indicated that the difference was present in both blocks, although the effect seemed reduced in half in the post-mix block [Pre-mix: $Estimate = 120.67$, $SE = 15.23$, $t = 7.92$, $p < .001$; Post-mix: $Estimate = 54.79$, $SE = 13.25$, $t = 4.13$, $p < .001$] (see Fig. 12).

3.3. Discussion

Nonbalanced bilinguals in Experiment 2 showed significant switching, mixing and post-mixing costs. Mixing costs were larger for the dominant than for the non-dominant language, consistent with a range of studies showing dominant deprioritization in mixing costs for non-balanced bilinguals (Christoffels et al., 2007; de Bruin et al., 2014; Gollan & Ferreira, 2009; Mosca & Clahsen, 2016; Peeters & Dijkstra, 2018). Switching costs did not differ between the two languages (i.e., were symmetrical), consistent with the usual finding in prior studies that only one marker between mixing and switching costs shows the asymmetry. Post-mixing costs showed the same asymmetrical pattern of dominant deprioritization as mixing costs. The dominant language was produced more quickly overall than the nondominant language, also consistent with prior literature (reviews in Hanulová et al., 2011; Runnqvist et al., 2011). In this experiment, repetition benefits were not statistically different between the two languages, but were numerically in the predicted direction (Kleinman & Gollan, 2018; Misra et al., 2012): on average, twice as large for the nondominant than for the dominant language, in both the pre-mix and post-mix blocks.

Further, the language difference was smaller at the end of the experiment than at the beginning at the expense of the dominant language English (another tendency also present, but weaker, for balanced bilinguals in Experiment 1). This pattern is another index of dominant deprioritization (although note that it is another way of stating that the post-mixing costs were larger for English than for Spanish).

In sum, Experiment 2 replicated the well-established dominant deprioritization effects for nonbalanced bilinguals (evident here in mixing costs). We also established robust dominant deprioritization effects in post-mixing costs. We next turn to monolinguals' within-language switching between basic-level names, which we consider dominant, and subordinate names, which we consider nondominant.

Table 5
LMER analyses of the data of Experiment 3.

Model	Predictors	Estimate	SE	t or z	p
Switching costs					
	Switching trial type (stay vs. switch)	30.41	7.72	3.94	.002
	Name type	-59.08	11.33	-5.22	< .001
	Switching trial type x Name type	18.30	9.79	1.87	.07
Mixing costs					
	Mixing trial type (pre-mix vs. mixed-stay)	190.71	19.58	9.74	< .001
	Name type	-56.91	15.07	-3.78	.001
	Mixing trial type x Name type	-23.81	16.86	-1.41	0.17
Post-mixing costs					
	Single naming block type (pre-mix vs. post-mix)	73.95	11.28	6.56	< .001
	Name type	-60.89	16.50	-3.69	.002
	Single naming block type x Name type	-31.58	15.24	-2.07	.04
Repetition benefits					
	Block type (single vs. mixed)	-187.92	15.64	-12.01	< .001
	Name type	-63.50	12.71	-5.00	< .001
	Repetition (single: 1-3; mixed: 1-6)	-13.94	2.33	-5.97	< .001
	Block type x Name type	-8.56	12.66	-.68	.50
	Block type x Repetition	-12.65	3.79	-3.34	.004
	Name type x Repetition	-4.34	2.62	-1.66	.11
	Block type x Name type x Repetition	-9.18	6.07	-1.51	.14

Note: Dark grey shading indicates significant effects, light grey shading indicates marginal effects.

4. Experiment 3: Name-type mixing in monolinguals

4.1. Method

4.1.1. Participants

Forty-eight functional monolinguals of American English, undergraduates at the University of California, San Diego (UCSD), participated for course credit. Language history characteristics are summarized in Table 1.

4.1.2. Materials

There were 12 colored photographs, four of which were the same as in the bilingual language-switching task (in bold below). The pictures were chosen to elicit both basic-level names (*bird*, *boat*, *book*, ***building***, *computer*, *dog*, *pants*, *flower*, *hat*, ***painting***, ***shoe***, ***tree***) and subordinate names (*owl*, *yacht*, *dictionary*, *skyscraper*, *laptop*, *Dalmatian*, *jeans*, *tulip*, *sombrero*, *portrait*, *sneaker*, *willow*). The basic-level names had higher frequency and were shorter than the subordinate names (SUBTLEX-US frequency, basic-level: $M = 78.33$, $SD = 55.28$; subordinate: $M = 3.30$,

$SD = 2.77$ [$t(11) = 4.68$, $p = .001$]; length in phonemes, basic-level: $M = 4.67$, $SD = 2.10$; subordinate: $M = 6.75$, $SD = 2.53$ [$t(11) = 2.38$, $p = .04$]).

4.1.3. Procedure

Subordinate names were elicited by requesting participants to “name the pictures in English using the most specific name you can think of, for example, *salmon* or *cantaloupe*”. Basic-level names were elicited by requesting participants to “name the pictures using a general, common name, for example, *fish* or *melon*”. The procedure was approved by the UCSD Institutional Review Board. All other aspects of the procedure were identical to the one in Experiments 1 and 2.

4.1.4. Design, coding and data analysis

These were the same as in Experiments 1 and 2. Analyses of naming latencies excluded 675 responses (4.9%) as voice-key inaccuracies, a further 190 responses (1.4%) as outliers and a further 350 responses (2.5%) as errors. Error rates are reported in Table 2.

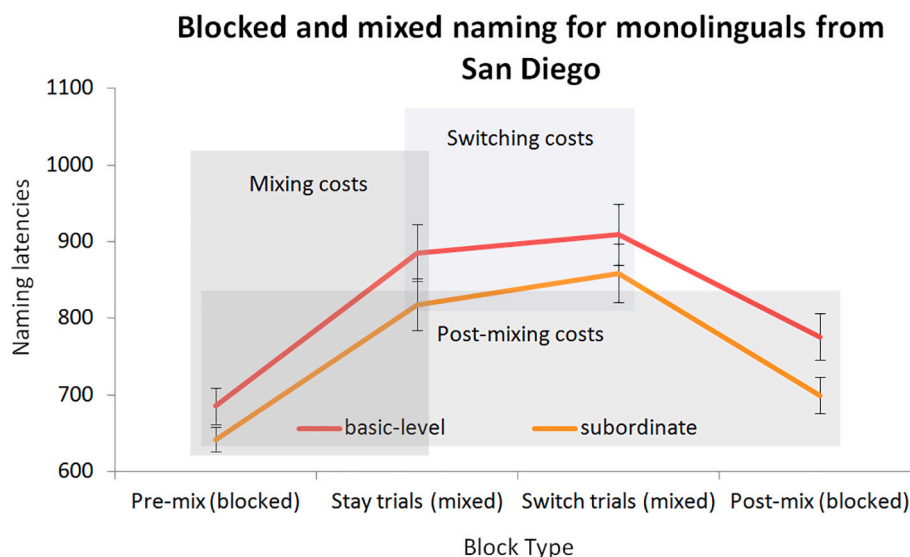


Fig. 7. By-participant mean naming latencies for each block type and name type for monolinguals in Experiment 3. Error bars represent 95% confidence intervals.

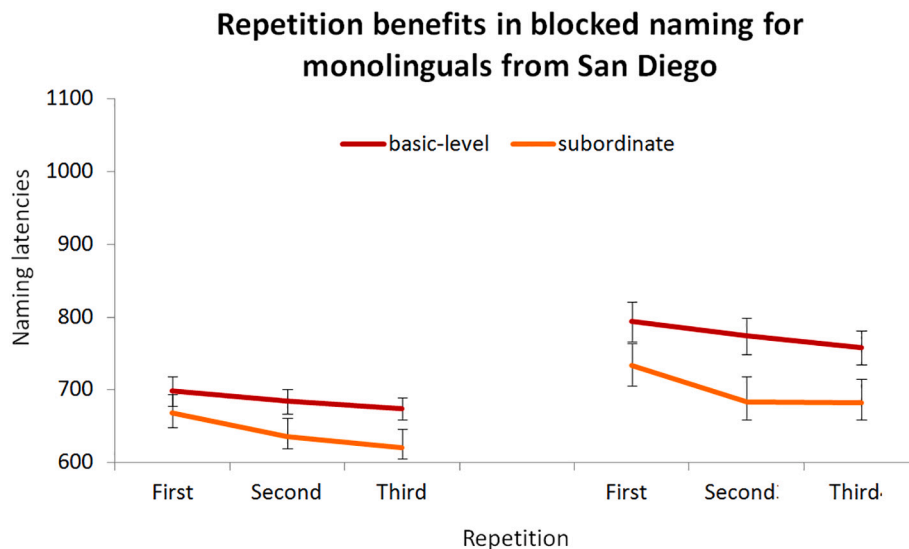


Fig. 8. By-participant mean naming latencies for each repetition in initial and final blocked naming for monolinguals in Experiment 3. Error bars represent 95% confidence intervals.

4.2. Results

The results of all statistical models are reported in Table 5. By-participant mean naming latencies are plotted in Fig. 7 (per block type and language) and Fig. 8 (per repetition in initial and final blocked naming).

4.2.1. Switching costs

Monolinguals switching between basic-level and subordinate naming incurred switching costs (the Switching trial type predictor in this analysis was significant). There was a tendency for larger costs for subordinate names than for basic-level names (there was a marginal interaction between Switching trial type and Name type).

4.2.2. Mixing costs

Monolinguals naming pictures with basic-level and subordinate names also incurred mixing costs: They named pictures on stay trials in mixed blocks more slowly than in the pre-mix blocks (the Mixing trial type predictor was significant). The mixing costs were also statistically equivalent for the two name types (the interaction between Mixing trial type and Name type was not significant).

4.2.3. Post-mixing costs

Monolinguals naming pictures with basic-level and subordinate names also incurred post-mixing costs (the Single naming block type predictor was significant). These costs were larger for basic-level than for subordinate names (the interaction between Single naming block type and Name Type was significant).

4.2.4. Repetition benefits

There were repetition benefits also in Experiment 3 (Repetition was a significant predictor), and these benefits were larger during blocked naming than during mixed naming (the interaction between Block type and Repetition was significant). Further, subordinate names seemed to benefit somewhat more from repetition than basic-level names (there was a marginal interaction between Name type and Repetition). As before, we ran additional models on the separate data of pre-mix, mixed and post-mix blocks with Name type, Repetition and their interaction as fixed predictors. These analyses showed significant repetition benefits in the pre-mix block [Repetition predictor: $Estimate = -19.03$, $SE = 4.58$, $t = -4.15$, $p = .001$], that were marginally larger for subordinate names than for basic-level names [Name type x Repetition interaction: $Estimate = -11.54$, $SE = 5.82$, $t = -1.98$, $p = .07$]. In the mixed and post-mix

blocks, there were also significant repetition benefits [Repetition predictor, mixed block: $Estimate = -7.57$, $SE = 2.12$, $t = -3.57$, $p = .001$; Repetition predictor, post-mix block: $Estimate = -21.61$, $SE = 6.24$, $t = -3.46$, $p = .004$], but those were similar for the two name types [Name type x Repetition interaction, $ps > 0.3$].

4.2.5. Name type dominance effects

All analyses showed significant effects of Name type. However, they reflected a reversed-dominance effect: Subordinate names were produced more quickly than basic-level names. Additional separate analyses on the data of the pre-mix and post-mix blocks, with Name type as the only fixed predictor, indicated that this was the case in both the pre-mix [Pre-mix: $Estimate = -45.05$, $SE = 20.82$, $t = -2.16$, $p = .04$] and post-mix blocks [$Estimate = -76.71$, $SE = 15.12$, $t = -5.07$, $p < .001$], but the difference became larger at the end of the experiment (see Fig. 12).

4.3. Discussion

In Experiment 3, monolinguals performed blocked naming and trial-level switching with pictures' basic-level and subordinate names. We assumed that basic-level and subordinate names' lexical representations present a representational imbalance within a language in some ways similar to that between dominant- and nondominant-language lexical representations for nonbalanced bilinguals. We were thus interested in signs of dominant deprioritization for basic-level names, as well as signs that basic-level names were really more dominant.

Indeed, monolinguals in Experiment 3 showed dominant deprioritization: Post-mixing costs were larger for basic-level names than for subordinate names. Mixing and switching costs were respectively equivalent for the two name types or slightly larger for subordinate names, and thus did not show dominant deprioritization.

Basic-level names accrued marginally smaller repetition benefits than subordinate names (cf. Griffin & Bock, 1998) in initial blocked naming, attesting to the representational imbalance of their lexical representations. Repetition benefits remained, but the asymmetry disappeared in subsequent blocks, presumably because of accessibility adjustments and control demands of the mixing context, as in the bilingual experiments.

Of note, monolingual within-language switching showed strong (symmetrical) repetition benefits in the mixed block, while for non-balanced bilinguals there were none. This difference may be related to the phonological switch entailed in between-language switching. The additional time necessary for it – for nonbalanced bilinguals, into and

Table 6
LMER analyses of the data of Experiment 4.

Model	Predictors	Estimate	SE	t or z	p
Switching costs					
	Switching trial type (stay vs. switch)	24.41	7.21	3.39	.006
	Name type	-48.31	12.15	-3.98	< .001
	Switching trial type x Name type	11.31	13.54	.84	.42
Mixing costs					
	Mixing trial type (pre-mix vs. mixed-stay)	201.42	21.73	9.27	< .001
	Name type	-29.86	14.46	-2.07	.05
	Mixing trial type x Name type	-48.83	17.04	-2.87	.007
Post-mixing costs on single naming					
	Single naming block type (pre-mix vs. post-mix)	82.65	17.54	4.71	< .001
	Name type	-27.02	15.07	-1.79	.09
	Single naming block type x Name type	-42.72	18.77	-2.28	.03
Repetition benefits					
	Block type (single vs. mixed)	-197.28	16.79	-11.75	< .001
	Name type	-41.58	13.36	-3.11	.006
	Repetition (single: 1-3; mixed: 1-6)	-17.33	3.15	-5.51	< .001
	Block type x Name type	1.70	10.86	.16	.88
	Block type x Repetition	-11.42	5.29	-2.16	.05
	Name type x Repetition	-13.49	3.50	-3.86	< .001
	Block type x Name type x Repetition	-12.83	6.49	-1.98	.05

Note: Dark grey shading indicates significant effects, light grey shading indicates marginal effects.

out of a less practiced phonological system – may make it impossible to accrue large repetition benefits during frequent switching; (cf. different effects of preparation time for between- and within-language switching in Declerck, Ivanova, et al., 2020).

Interestingly, unlike any of the bilingual groups, monolinguals in Experiment 3 showed complete name dominance reversal: Across the whole experiment, nondominant names were produced faster than dominant names. This result was unexpected: No theory of bilingual or monolingual lexical access or piece of empirical evidence would predict that, everything else being equal, lower-frequency longer words would be produced faster than higher-frequency shorter words. For this reason, we began suspecting that this dominance reversal was not in its core related to representation or control mechanisms but to the fact that some of the subordinate names were better labels for the pictures we used to elicit them than basic-level names. Consistent with this possibility, Declerck, Ivanova, et al. (2020) did not find a dominance reversal when bilinguals switched between neutral-register words corresponding to our basic-level names (e.g., *dog*), and informal-register words (e.g., *pooch*): The latter were produced significantly more slowly overall. Important here, this result suggests that dominance reversal does not universally result when two name types of different frequency are mixed together within a language.

In any case, we find it noteworthy that basic-level names were produced more slowly than subordinate names, yet they initially accrued somewhat smaller repetition benefits and eventually suffered more from mixing. These patterns offer further support to the possibility that the dominance reversal was unrelated to representational robustness or control. It may suggest that the language system assesses the potential for competition (and, from it, the strength of the control it needs to apply) at least in part a priori, on the basis of general name accessibility (e.g., context-independent resting activation) and not on the basis of context-specific name appropriateness (i.e., context-driven temporary activation).

Before interpreting the monolingual pattern of results, we conducted Experiment 4, with the following purpose. First, this experiment aimed to see if the pattern of Experiment 3 would replicate. Second, it tested monolinguals from the same community as the bilinguals in Experiments 1 and 2. Third, it replaced the basic-level names for two pictures with more suitable ones, as an attempt to reduce the reversed-dominance effect. (We did not replace more items at this point because we only realized the full extent of the issue after collecting production rates following completion of Experiment 4, see below.)

5. Experiment 4 (replication): Name-type mixing in another group of monolinguals

5.1. Method

5.1.1. Participants

Forty-eight functional monolinguals of American English, undergraduates at the University of Texas at El Paso, participated for course credit or pay (\$10/h). One additional participant, who had a large Spanish vocabulary, was excluded from analyses. Participants' characteristics are summarized in Table 1.

5.1.2. Materials

The materials were the same as in Experiment 3, with the following exceptions: *jeans* was replaced by *dress*, and a sparrow (instead of owl) photograph was selected for *bird*.

5.1.3. Procedure, design, coding and data analysis

These were the same as in Experiments 1 and 2. Analyses of naming latencies excluded 993 responses (7.2%) as voice-key inaccuracies, a further 216 responses (1.7%) as outliers and a further 343 responses (2.5%) as errors. Error rates are reported in Table 2.

5.2. Results

The results of all statistical models are reported in Table 6. By-participant mean naming latencies are plotted in Fig. 9 (per block type and language) and Fig. 10 (per repetition in initial and final blocked naming).

5.2.1. Switching costs

Monolinguals in Experiment 4 incurred switching costs that were symmetrical across the two name types (the Switching trial type predictor was significant, and there was no interaction between Switching trial type and Name type).

5.2.2. Mixing costs

Monolinguals in Experiment 4 also incurred mixing costs (the Mixing trial type predictor in this analysis was significant). These costs were larger for basic-level names than for subordinate names, that is, showed dominant deprioritization (the interaction between Mixing trial type and Name type was significant).

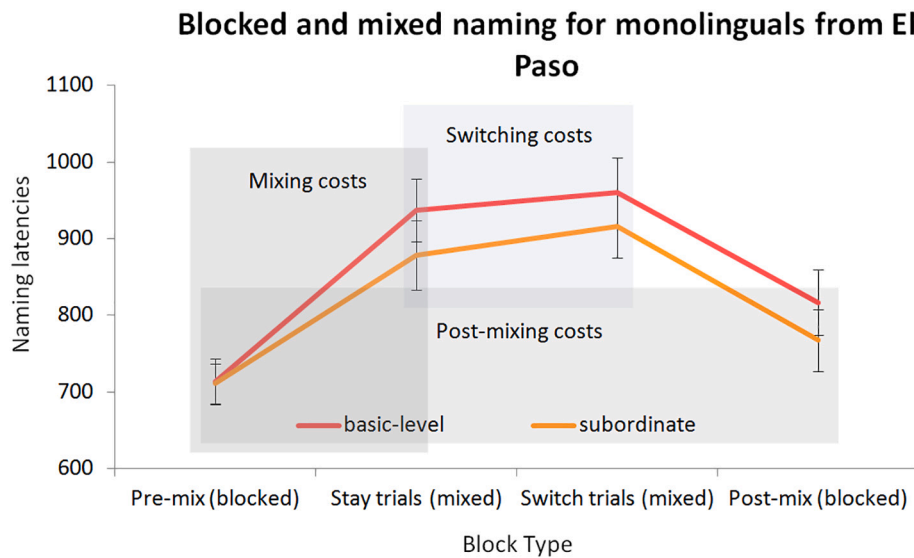


Fig. 9. By-participant mean naming latencies for each block type and name type for monolinguals in Experiment 4. Error bars represent 95% confidence intervals.

5.2.3. Post-mixing costs

Monolinguals in Experiment 4 also named pictures more slowly during final blocked naming than during initial blocked naming (the Single naming block type predictor was significant). These costs also showed dominant deprioritization: They were larger for dominant (basic-level) names than for nondominant (subordinate) names (the interaction between Single naming block type and Name type was significant).

5.2.4. Repetition benefits

As in all three previous experiments, naming latencies in Experiment 4 showed repetition benefits (Repetition was a significant predictor), and these benefits were larger in single-naming blocks than in mixed blocks (the interaction between Block type and Repetition was significant). Further, the significant interaction between Name type and Repetition indicated larger repetition benefits for subordinate than for basic-level names. This effect was in the same direction – but stronger – as for the monolinguals in Experiment 3. The three-way interaction between Repetition, Block type and Name type was also significant. To shed light on these patterns, we ran additional models on the separate

data of pre-mix, mixed and post-mix blocks, with the fixed predictors Repetition, Name type, and their interaction. In the pre-mix block, there were significant repetition benefits [Repetition predictor: $Estimate = -29.58$, $SE = 6.47$, $t = -4.57$, $p < .001$] that were larger for subordinate names than for basic-level names [Repetition x Language interaction: $Estimate = -21.23$, $SE = 8.23$, $t = -2.58$, $p = .03$]. In the mixed block, there were also repetition benefits [Repetition predictor: $Estimate = -11.48$, $SE = 2.81$, $t = -4.08$, $p < .001$] that were also larger for subordinate names than for basic-level names [Name type x Repetition interaction: $Estimate = -7.24$, $SE = 3.02$, $t = -2.40$, $p = .02$]. In the post-mix block, there were no overall repetition benefits [Repetition predictor, $p = .11$], but the benefits were marginally larger for subordinate than for basic-level names [Repetition x Language interaction: $Estimate = -18.39$, $SE = 8.57$, $t = -2.15$, $p = .06$].

5.2.5. Name type dominance effects

The effect of Name type was significant in three of the reported analyses and marginal in the fourth one: Overall, monolinguals in Experiment 4 produced subordinate names more quickly than basic-level names. This was the same pattern of dominance reversal (numerically

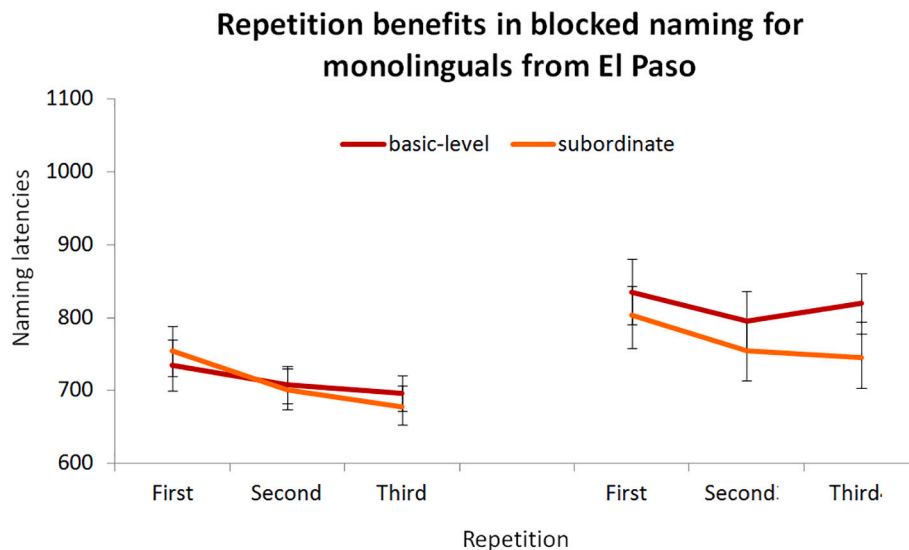


Fig. 10. By-participant mean naming latencies for each repetition in initial and final blocked naming for monolinguals in Experiment 4. Error bars represent 95% confidence intervals.

somewhat weaker) as shown by monolinguals in Experiment 3. Additional light on this pattern was shed by the follow-up separate analyses on the data of the pre-mix and post-mix single-naming blocks (the models had Name type as the only fixed predictor). Subordinate names were produced more quickly in the post-mix single naming block [*Estimate* = -48.49, *SE* = 16.83, *t* = -2.88, *p* = .009] but in the pre-mix block they were produced with a similar speed to basic-level names [*Estimate* = -5.80, *SE* = 18.64, *t* = -0.31, *p* = .76].

5.3. Discussion

In Experiment 4, monolinguals mixing basic-level and subordinate names incurred mixing, switching, and post-mixing costs (as participants in all previous experiments). The mixing and post-mixing costs were larger for basic-level names than for subordinate names. As before, we interpret the costs as evidence for dominant deprioritization, here of basic-level names in favor of subordinate names.

Supporting the pattern of dominance, there were also repetition benefits that were smaller for basic-level names than for subordinate names throughout the whole experiment. This asymmetry was overall predicted (Griffin & Bock, 1998) and supports our assumption that basic-level names have more robust lexical representations than subordinate names (and are thus the ones that need to be deprioritized more strongly). The repetition benefits asymmetry was stronger than that observed in all other experiments (where it emerged only in the beginning of the experiment). Again, we note that such an asymmetry emerged (and, here, remained stable) in the presence of the overall speed advantage of the more weakly represented names. As before, we interpret this pattern as evidence for different effects of baseline representational robustness and temporary activation.

Experiment 4 also replicated Experiment 3's name dominance reversal (in a slightly attenuated form, possibly because of having replaced two items). To see if the reversal was related to greater name suitability of subordinate names, we collected production rates for the images used in Experiment 4 by asking a different group of monolingual participants (final *N* = 29) to type the first name that came to mind for each image. Eight of the 12 images elicited basic-level names from fewer than half of the participants (range: 10%–48%). Crucially, there was a strong relationship between the suitability of basic-level names and the size of name dominance reversal in Experiment 4: the higher the suitability, the smaller the reversal (see Fig. 13 in Appendix B). This was supported by a large Pearson bivariate correlation ($r^2(10) = 0.64$, *p* = .03) and by a significant interaction between Name type (reflecting dominance effects) and Basic-level names' production rates in a LMER analysis [*p* = .03]. These analyses (reported in full in Appendix B) point to name suitability rather than representation or control processes as the cause of the name dominance reversal in Experiments 3 and 4.

6. Additional analyses

6.1. Revisiting the dominance of basic-level names

We interpreted our findings as dominant deprioritization for both nonbalanced bilinguals and monolinguals, under the assumption that basic-level names are dominant for monolinguals. However, basic-level names showed a dominance reversal, which raises the question if this assumption is warranted in the current context. We think it is, for two reasons. First, basic-level names showed smaller repetition benefits, as predicted; we are not aware of any theory or mechanism predicting larger repetition benefits for dominant names. Second, basic-level names did show a regular dominance effect (that is, were produced more quickly than subordinate names) the very first time they were produced by monolinguals who started with them [main effect of Name type for the first repetition of the pre-mix blocks for those monolinguals across both experiments who named basic-level names first: *Estimate* = 50.87, *SE* = 22.76, *t* = 2.24, *p* = .04]. In other words, even if basic-level names

were overall somewhat less preferred for our stimuli, they were still named faster before repetition and mixing effects kicked in. At the same time, looking at the first repetition only, these names still incurred larger post-mixing costs than subordinate names for these participants, that is, showed deprioritization [Single naming block type x Name type interaction: *Estimate* = -133.34, *SE* = 27.83, *t* = -4.79, *p* < .001; neither of these effects differed across experiments as there were no interactions with the Monolingual experiment predictor, all *ps* > 0.2]. We thus keep the assumption that basic-level names were representationally dominant, and at least somewhat similar in that respect to dominant-language names for bilinguals.

6.2. Comparison of control markers within and across languages

We consider that the presence of dominant deprioritization in monolinguals constitutes the main theoretical contribution of this study, but we also conducted follow-up analyses (described in detail in Appendix C) to see how the deprioritization magnitudes in monolinguals compare to those of nonbalanced bilinguals (and how those of the two bilingual groups compared to one another). In the analyses we conducted, the statistical indicator of whether a type of cost differed between the two groups is the three-way interaction between Trial or Block type, Name type and Language group. The *p*-values for these interactions in each of three analyses (with all or only the shared items across experiments), for each of the three cost types (switching, mixing and post-mixing) are reported in Table 7. Switching, mixing and post-mixing costs in all experiments are respectively plotted in Panels A, B and C of Fig. 11. The advantage gained by nondominant over dominant names from initial to final blocked naming in all experiments is plotted in Fig. 12.

We did not have a priori predictions about these between-experiment comparisons, beyond the expectation that such magnitudes would differ more clearly between the nonbalanced and balanced bilinguals (where they are theoretically predicted, even if the difference is one of degree) than between the nonbalanced bilinguals and monolinguals. This expectation was supported. An examination of Table 7 suggests that all three types of cost were significantly different between nonbalanced and balanced bilinguals, while the difference between the switching and post-mixing costs between nonbalanced bilinguals and monolinguals was not significant. The mixing cost asymmetry was significantly larger for nonbalanced bilinguals (87 ms) than for monolinguals (39 ms). This could be because the representational imbalance may on average be smaller between representations within a single language than between dominant and nondominant language representations (because of the variability in language dominance among nonbalanced bilinguals).

7. General discussion

This study asks if monolinguals wanting to ensure they say the appropriate word in the presence of highly activated within-language competitors can use a similar mechanism to those used by bilinguals to ensure they speak the appropriate language. Regardless of which theory is adopted to explain bilingual language control, such control is behaviorally signaled by dominant deprioritization: Reduced speed, accessibility or accuracy of the dominant language in a dual-language context relative to a single-language context. The empirical question of main interest in this study was thus whether monolinguals mixing dominant (basic-level) and nondominant (subordinate) names would also show any signs of dominant deprioritization.

Balanced (Experiment 1) and non-balanced bilinguals (Experiment 2) named pictures in two languages (in single-language and mixed-language blocks) and monolinguals (Experiments 3–4) named pictures with basic-level and subordinate names (in identically-structured single and mixed blocks). We explored several established bilingual indicators of language control: Mixing and switching costs and dominance effects. We also explored a type of cost previously unexplored as such, post-mixing costs (the difference between single-naming speed before and

Table 7

P-values for interactions reflecting control marker comparisons in between-experiment LMER comparisons of nonbalanced bilinguals with monolinguals and balanced bilinguals.

Nonbalanced bilinguals (Exp. 2) compared to:		
	Monolinguals (Exp. 3-4 pooled, all items/ Exp. 3-4 pooled, 4 shared items / Exp. 4, 6 shared items)	Balanced bilinguals (Exp. 1; all items are shared)
Switching costs (Switching trial type x Name type x Language group)	$p = .95 / p = .43 / p = .48$	$p = .05$
Mixing costs (Mixing trial type x Name type x Language group)	$p = .03 / p = .41 / p = .49$	$p < .001$
Post-mixing costs (Single naming block type x Name type x Language group)	$p = .24 / p = .48 / p = .96$	$p = .04$
	Monolinguals Exp. 3 (all items / 4 shared items)	Monolinguals Exp. 4 (all items / 6 shared items)
Repetition benefits (Name type x Repetition x Language group)	$p = .73 / p = .86$	$p = .04 / p = .17$

Note: Dark grey shading indicates significant effects, light grey shading indicates marginal effects.

after trial-level switching). Finally, we explored repetition benefits as a marker of representational robustness (less robustly represented words accrue larger benefits).

We found indices of dominant deprioritization in all experiments. For nonbalanced bilinguals, this was evident in mixing and post-mixing costs. For balanced bilinguals (whose English seemed slightly more dominant), this was still evident in post-mixing costs. Most noteworthy, monolinguals also showed dominant deprioritization during within-language switching: in post-mixing costs (in both experiments) and mixing costs (in Experiment 4). In follow-up analyses, the magnitude of dominant deprioritization seemed to differ more clearly between nonbalanced and balanced bilinguals (as predicted *a priori*),⁷ than between nonbalanced bilinguals and monolinguals. We take these results to suggest that monolinguals are able to use functionally similar mechanisms to avoid interference from highly activated same-language competitors to those used by bilinguals to avoid interference from between-language competitors.

What could be the nature of such shared control mechanisms? Most authors assume that bilingual control mechanisms are top-down (e.g., Blanco-Elorrieta & Pykkänen, 2016; Green, 1998; Seo et al., 2018). Monolingual interference resolution, on the other hand, has been attributed to bottom-up mechanisms - lateral inhibition or attaining critical activation difference with competitors. Here we propose that the shared control mechanisms we investigated have a top-down component in both bilinguals and monolinguals. In other words, a top-down intention to use particular words (in a particular language), possibly together with a signal for increased interference from a monitoring system, would trigger goal-directed activation (Philipp et al., 2007) or inhibition of representations (Green, 1998) or changes of activation thresholds (Branzi et al., 2014).⁸

This proposal, awaiting direct testing in future research, is motivated

⁷ We do not mean to imply that balanced bilinguals use a different *kind* of control mechanism than non-balanced bilinguals. A possible reason for different patterns of control indices is that control is applied proportionally to representational strength (e.g., Green, 1998), which differs between balanced and nonbalanced bilinguals for one of their languages. It could be that this alone accounts for the different patterns. On the other hand, it is possible that balanced and non-balanced bilinguals use categorically different mechanisms for between-language interference resolution (cf. Costa et al., 2006).

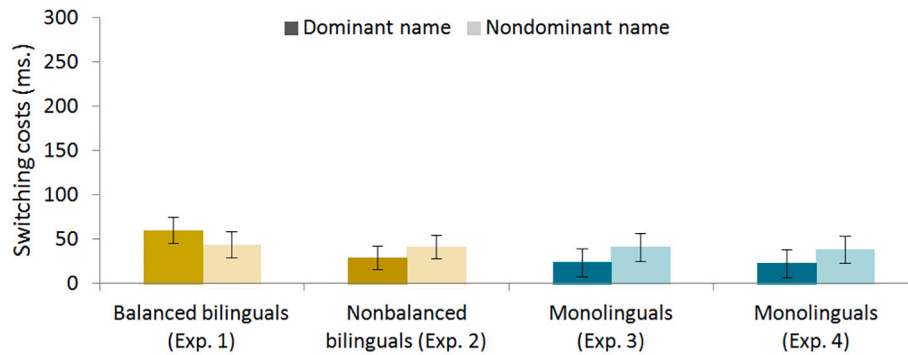
⁸ We note that a “mental firewall” between bilinguals’ two languages (Costa et al., 1999) seems inconsistent with the current findings because it does not straightforwardly account for deprioritization effects.

by the reasoning that most bottom-up mechanisms do not seem able to explain dominant deprioritization, and no bottom-up mechanism by itself seems to account for the findings of this study. Both the (automatic) activation and lateral inhibition spreading from one representation to another should be proportional to the source representation’s activation (e.g., Levelt et al., 1999; also consistent with Dylman & Barry’s, 2018, proposal of weaker production connections from nondominant- to dominant-language representations, or from less frequent to more frequent synonyms, than vice versa). If we assume that the names with higher activation were the ones produced more quickly in our experiments, these would be dominant-language names in the bilingual experiments, and subordinate names in the monolingual experiments. It then becomes unclear why dominant-language names patterned instead with basic-level names (showing larger costs of mixing), and subordinate names patterned with nondominant-language names (showing larger repetition benefits).

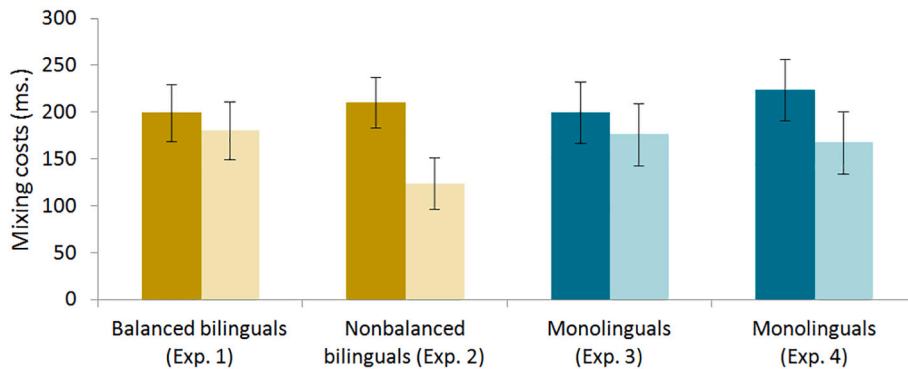
Is it possible to explain the current results without lexical competition? As mentioned in the Introduction, the response-exclusion account could in principle explain dominant deprioritization: In demanding, high-conflict situations, the language system subjects to an extra careful examination dominant responses because they reach the articulatory buffer too fast, and this ultimately leads to their disproportional delay. However, this explanation holds only when dominant responses have an overall speed advantage. As such, it would remain unclear why, for both monolingual groups tested here, basic-level names incurred larger costs – given that they were the overall slower responses. For the same reasons, Oppenheim, Dell, and Schwartz’s (2010) incremental learning model also seems to need additional assumptions to explain the full results reported here.

We cannot completely discard the possibility that different mechanisms explain bilingual and monolingual performance in our study: In principle, similar behavior may not necessarily result from the same underlying mechanisms. But we think this possibility is unlikely, for several reasons. First, to the extent that within-language lexical selection in monolinguals and between-language lexical selection in bilinguals happen in the same type of system (i.e., have the same core features and are governed by the same underlying principles, e.g., spreading activation), it is not straightforward to assume that similar behavior would result from radically different underlying mechanisms. Second, as we point out above, of the two different mechanisms that have been put forward to explain lexical dynamics in such systems (top-down versus bottom-up), only one is able to explain all current results (together with other results from the bilingual literature). We thus conclude in favor of shared top-down mechanisms, and hope that future research will shed

Panel A: Switching costs in all experiments



Panel B: Mixing costs in all experiments



Panel C: Post-mixing costs in all experiments

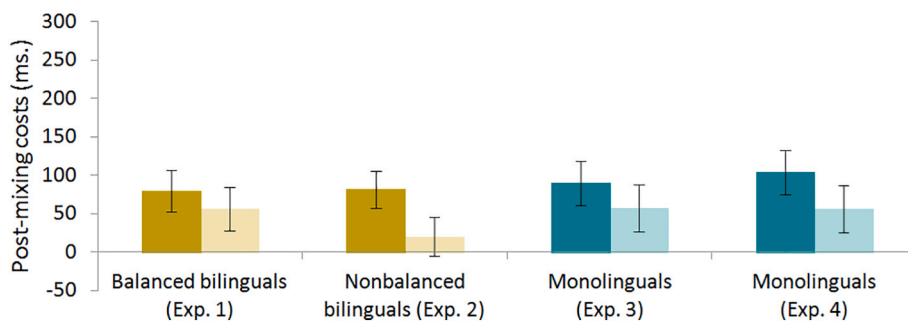


Fig. 11. Switching, mixing and post-mixing costs in all experiments. Panel A: Switching costs. Panel B: Mixing costs. Panel C: Post-mixing costs.

more light on this issue.

When would monolinguals use such shared control mechanisms? While this study does not provide direct evidence, restricting oneself to a particular register (e.g., saying *nice*) seems likely to necessitate control mechanisms to prevent interference from frequently used words belonging to a different register (e.g., *cool*). Another such situation could be a consecutive change of conversation partners each of which uses a different name for the same entity. Speakers establish *conceptual pacts* of how to refer to entities with each person they converse with (Brennan & Clark, 1996); *couple speak* is a notable example. That is, if a speaker refers to a dog as *puppy wuppy* or to a treat as *tort* with their significant other (examples from Weiss, 2018), they would have these names highly activated but infelicitous for an immediately following conversation with a non-initiated (and likely mocking) friend. The current results suggest that the more robustly represented of two names would be the one deprioritized, regardless of the level of temporary activation in any conversation. We note, however, that the control mechanisms studied here would be

used more frequently by bilinguals than by monolinguals, because bilinguals need to prevent both within- and between-language interference. Also, between-language and within-language interference resolution may not necessitate the use of such mechanisms equally often.

7.1. Other implications

Bilingual language control can be local (over translation equivalents) or global (over a whole language; Guo, Liu, Misra, & Kroll, 2011; Branzi, Della Rosa, Canini, Costa, & Abutalebi, 2016). The current evidence of within-language control in monolinguals concerns local control, and our study does not provide evidence about global control. In fact, the current results can be explained without the notions of language or category membership (e.g., language tags): It seems possible that competition between two highly activated representations can be resolved without regard for the nature or category membership of these representations. In any case, for future work we hypothesize that global control can also

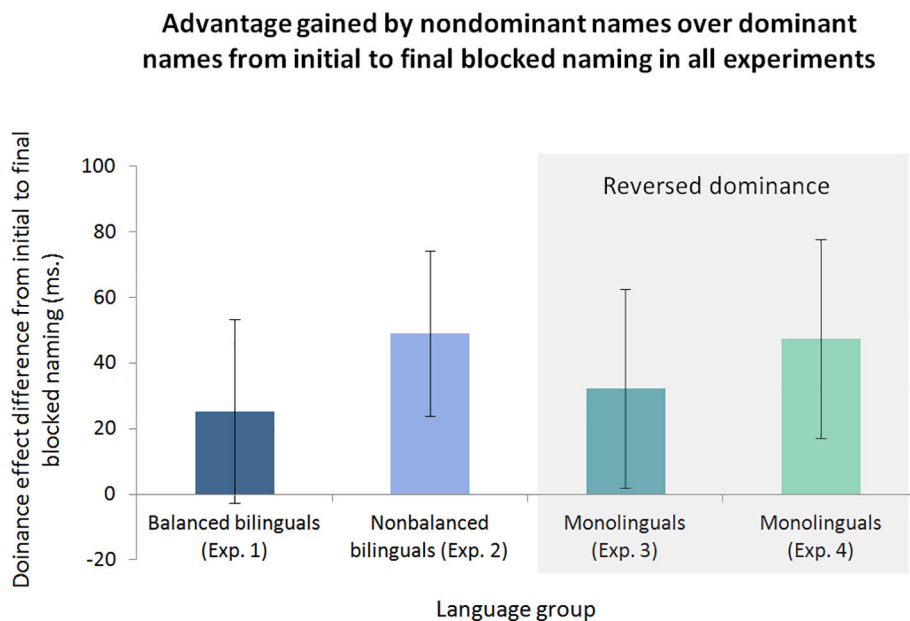


Fig. 12. Magnitude of the advantage gained by nondominant names over dominant names from initial to final blocked naming in all experiments. It was calculated as (dominant minus nondominant naming latencies in the pre-mix blocks) minus (dominant minus nondominant naming latencies in the post-mix blocks).

be applied during monolingual within-language interference resolution but has a more limited scope; specifically, it may be restricted by available within-language word categories (e.g., different registers) or individuals' capacity to create ad-hoc categories.

A robust difference between monolinguals and bilinguals in the current study was the presence of standard dominance effects in bilinguals (dominant-language names were produced more quickly overall than nondominant-language names) but reversed dominance effects in monolinguals (basic-level names were produced more slowly than subordinate names). A negative relationship between independently-elicited production rates of basic-level names for the experimental images, on the one hand, and the magnitudes of dominance reversal for each image in Experiment 4, on the other, suggested name suitability as a factor behind the monolingual dominance reversal: the less likely an image was to elicit the basic-level name, the larger the dominance reversal. We take this to imply that the overall monolingual dominance reversal here was largely driven by a process unrelated to language control, and unrelated to processes underlying dominance reversal attested in prior bilingual experiments (e.g., Costa et al., 2006; Costa & Santesteban, 2004; Li & Gollan, 2018). However, we contend that the magnitude of the reversal did increase over the different blocks because of control processes. Of note, we found standard dominance effects when basic-level names were produced in the very first block for the very first time – that is, before the larger mixing costs for basic-level names and the larger repetition benefits for subordinate names had kicked in.

Switching costs in our experiments did not show dominant deprioritization, consistent with prior studies looking at both mixing and switching costs: An asymmetry tends to appear in one of the two markers but not both (e.g., Christoffels et al., 2007; Declerck et al., 2013; Gollan & Ferreira, 2009; Mosca & Clahsen, 2016). Which marker shows the deprioritization may depend on study specifics, but such results suggest that mixing and switching costs at least in part reflect a process that is shared between the two. It also suggests that dominant deprioritization has a ceiling: The cost asymmetry does not accrue beyond a certain point. In our study, this was the case not only for dominant-language names for bilinguals, but also for basic-level names for monolinguals.

A marker that consistently showed dominant deprioritization in all four experiments was post-mixing costs, leading to a reduction or reversal of the dominance effect. We assume that these costs reflect sustained interference effects between two highly activated names with

near-alternating appropriateness as responses in the previous block, together with a deprioritization mechanism to diminish the interference from responses considered dominant by the system. Post-mixing costs may also reflect persisting heightened monitoring (e.g., careful examination of the appropriateness of each response) after a period of trial-level switching and thus may be lower or convert into benefits in studies that only involve blocked naming or a large number of naming trials (cf. Christoffels et al., 2016). In any case, persistent effects of language mixing (measured here by post-mixing costs) are consistent with longer-term deprioritization effects on the dominant language found after a period of language immersion (Baus, Costa, & Carreiras, 2013; Linck, Kroll, & Sunderman, 2009). The contribution of our study is that post-mixing costs can be used as another index of dominant deprioritization, including between synonyms within a language.

Repetition benefits were larger for subordinate and nondominant-language names in the first two single-naming blocks, also consistent with prior studies (Griffin & Bock, 1998; Kleinman & Gollan, 2018; Misra et al., 2012). The current results contribute to the literature on repetition benefits and repetition priming by showing that the magnitude of the benefit depends on overall representational robustness (related to e.g. frequency), not transient activation (related to e.g., name appropriateness).

8. Conclusion

We found that monolinguals naming pictures with basic-level and subordinate names showed dominant deprioritization, a similar pattern to that of nonbalanced bilinguals naming pictures in their dominant and nondominant language. As predicted, nondominant-language names (in initial blocked naming) incurred larger repetition benefits than dominant-language names (in Experiment 2, numerically). Subordinate names also incurred larger repetition benefits than basic-level names (in initial blocked naming in Experiment 3 or in the whole Experiment 4). This was so even though subordinate names were produced overall faster than basic-level names (except when basic-level names were produced for the first time in the very first single-language block). On the basis of these results, we conclude that a functionally similar mechanism can manage interference both between and within languages. Such a mechanism involves deprioritization of names with more robust lexical representations regardless of their transient, context-

driven activation. We propose that such a mechanism is top-down, as the mechanisms proposed to explain how bilinguals resolve between-language lexical interference.

CRedit authorship contribution statement

Iva Ivanova: Conceptualization, Methodology, Data curation, Formal analysis, Visualization, Writing - original draft, Writing - review & editing. **Dacia Carolina Hernandez:** Project administration, data coding.

Acknowledgements

Many thanks to Yashna Bowen, Coy Shaffstall, Mayra Murrillo,

Kimberly Esquivel-Uriar and Sharon Chee at UCSD, and Bridgette Leyva, Alyssa Cardona, Victoria Jeffress, Lluvia Mendiola, Aziz Atiya, Elisa Espinoza and Jonathan Ramirez at UTEP, for help with data collection, and to Tamar Gollan, Vic Ferreira, and the members of the Ferreira-Gollan Language Production Lab for helpful discussions. Experiment 3 was conducted during I.I.'s postdoctoral fellowship at UCSD, which was funded by an R01 from NICHD (HD050287) and an R01 from NIHCD (DC011492), awarded to Tamar H. Gollan, and by an R01 from NICHD (HD051030), awarded to Victor S. Ferreira. The authors have no competing interests to declare.

Appendix A. Analyses of error rates

Table 8

LMER analyses of error rates in Experiment 1.

Model	Predictors	Estimate	SE	z	p
Switching costs					
	Switching trial type (stay vs. switch)	.43	.13	3.37	< .001
	Name type	.18	.21	.84	.40
	Switching trial type x Name type	.42	.25	1.70	.09
Mixing costs					
	Mixing trial type (pre-mix vs. mixed-stay)	.08	.17	.52	.61
	Name type	.25	.24	1.04	.30
	Mixing trial type x Name type	-.58	.37	-1.56	.12
Post-mixing costs on single naming					
	Single naming block type (pre-mix vs. post-mix)	-1.43	.25	-5.60	< .001
	Name type	.46	.48	.95	.34
	Single naming block type x Name type	-.36	.46	-.79	.43
Repetition benefits					
	Block type (single vs. mixed)	-1.30	.19	-6.80	< .001
	Name type	.09	.26	.36	.72
	Repetition (single: 1-3; mixed: 1-6)	-.30	.06	-4.99	< .001
	Block type x Name type	-.34	.31	-1.08	.28
	Block type x Repetition	-.23	.12	-1.93	.054
	Name type x Repetition	-.33	.12	-2.79	.005
	Block type x Name type x Repetition	-.31	.24	-1.26	.21

Table 9

LMER analyses of error rates in Experiment 2.

Model	Predictors	Estimate	SE	z	p
Switching costs					
	Switching trial type (stay vs. switch)	.23	.13	1.75	.08
	Name type	.46	.29	1.59	.11
	Switching trial type x Name type	.07	.21	.38	.71
Mixing costs					
	Mixing trial type (pre-mix vs. mixed-stay)	-.08	.17	-.52	.60
	Name type	.69	.40	1.72	.09
	Mixing trial type x Name type	-.90	.30	-2.98	.002
Post-mixing costs on single naming					
	Single naming block type (pre-mix vs. post-mix)	-1.04	.22	-4.78	< .001
	Name type	1.04	.51	2.05	.04
	Single naming block type x Name type	-.49	.48	-1.01	.31
Repetition benefits					
	Block type (single vs. mixed)	-.60	.13	-4.61	< .001
	Name type	.64	.35	1.82	.07
	Repetition (single: 1-3; mixed: 1-6)	-.18	.05	-3.50	< .001
	Block type x Name type	.59	.27	2.20	.03
	Block type x Repetition	-.09	.10	-.94	.35
	Name type x Repetition	.03	.08	.40	.69
	Block type x Name type x Repetition	.05	.17	-.29	.77

Table 10

LMER analyses of error rates in Experiment 3.

Model	Predictors	Estimate	SE	z	p
Switching costs					
	Switching trial type (stay vs. switch)	.44	.14	3.22	.001
	Name type	-.31	.17	-1.84	.07
	Switching trial type x Name type	.45	.27	1.66	.10
Mixing costs					
	Mixing trial type (pre-mix vs. mixed-stay)	.28	.18	1.59	.11
	Name type	-.41	.29	-.29	.16
	Mixing trial type x Name type	.30	.41	-.73	.47
Post-mixing costs on single naming					
	Single naming block type (pre-mix vs. post-mix)	-.67	.29	-2.32	.02
	Name type	-.57	.32	-1.77	.08
	Single naming block type x Name type	-.65	.67	-.97	.33
Repetition benefits					
	Block type (single vs. mixed)	-1.02	.17	-6.02	< .001
	Name type	-.31	.19	-1.64	.10
	Repetition (single: 1-3; mixed: 1-6)	-.20	.06	-3.14	< .001
	Block type x Name type	-.07	.35	-.21	.83
	Block type x Repetition	-.10	.14	-.72	.47
	Name type x Repetition	.08	.12	.63	.53
	Block type x Name type x Repetition	.25	.30	.83	.41

Table 11

LMER analyses of error rates in Experiment 4.

Model	Predictors	Estimate	SE	z	p
Switching costs					
	Switching trial type (stay vs. switch)	.30	.17	1.75	.08
	Name type	-.21	.18	-1.17	.24
	Switching trial type x Name type	.46	.29	1.58	.11
Mixing costs					
	Mixing trial type (pre-mix vs. mixed-stay)	.40	.31	1.31	.19
	Name type	.10	.35	.27	.79
	Mixing trial type x Name type	-1.27	.70	-1.82	.07
Post-mixing costs on single naming					
	Single naming block type (pre-mix vs. post-mix)	-.08	.39	-.20	.84
	Name type	.16	.42	.37	.71
	Single naming block type x Name type	-.88	.73	-1.22	.22
Repetition benefits					
	Block type (single vs. mixed)	-.73	.18	-4.02	< .001
	Name type	.04	.25	.14	.89
	Repetition (single: 1-3; mixed: 1-6)	-.19	.07	-2.66	< .001
	Block type x Name type	.72	.45	1.59	.11
	Block type x Repetition	-.10	.14	-.73	.47
	Name type x Repetition	.02	.11	.15	.88
	Block type x Name type x Repetition	-.05	.25	.21	.83

Appendix B. Items in Experiment 4: Basic-level name production rates and relationship with dominance reversal

Production rates were collected from 54 additional participants (students at the University of Texas at El Paso). Of them, 25 participants were excluded because they reported exposure to a language other than English in the first seven years of life, or using English less than 90% per day; final production rates were calculated for 29 participants (but the patterns were similar when rates were calculated from all participants). Participants typed in an online form the first name that came to mind for the twelve photographs used in Experiment 4.

An examination of Fig. 13 confirms that eight of the 12 images used in Experiment 4 (all of which were also used in Experiment 3) more readily elicited subordinate names than basic-level names, and that there was a strong relationship between the suitability of basic-level names for the experimental images and the size of dominance reversal in Experiment 4.

To look for statistical support of this relationship, we conducted two analyses. First, we found a strong Pearson bivariate correlation between the percentage basic-level names produced and the size of dominance reversal ($r^2(10) = 0.64, p = .03$): the greater the preference for a basic-level name, the smaller the dominance reversal (i.e., the smaller the advantage the subordinate name had over the basic-level name in Experiment 4).

Second, we ran an LMER model with the fixed predictors Name type (coded as above), Basic-level names' production rates (a continuous predictor centered around the mean), and their interaction. There was a significant interaction between Name type and Basic-level names' production rates [$Estimate = 1.05, SE = 0.42, t = 2.50, p = .03$], reflecting the fact that the dominance reversal was smaller for items that more frequently elicited the basic-level name. This result supports our hypothesis that the overall dominance reversal we observed for monolinguals switching between basic-level and subordinate names was more likely related to stimulus properties (and less likely to different control mechanisms). However, we take the fact that the size of the reversal became larger throughout the experiment as an index of dominant deprioritization (i.e., as the result of control mechanisms). This gradual enlargement is also why Fig. 13 shows a reversal even for items biased towards the basic-level name (e.g., *bird-sparrow* and *dress-gown*).

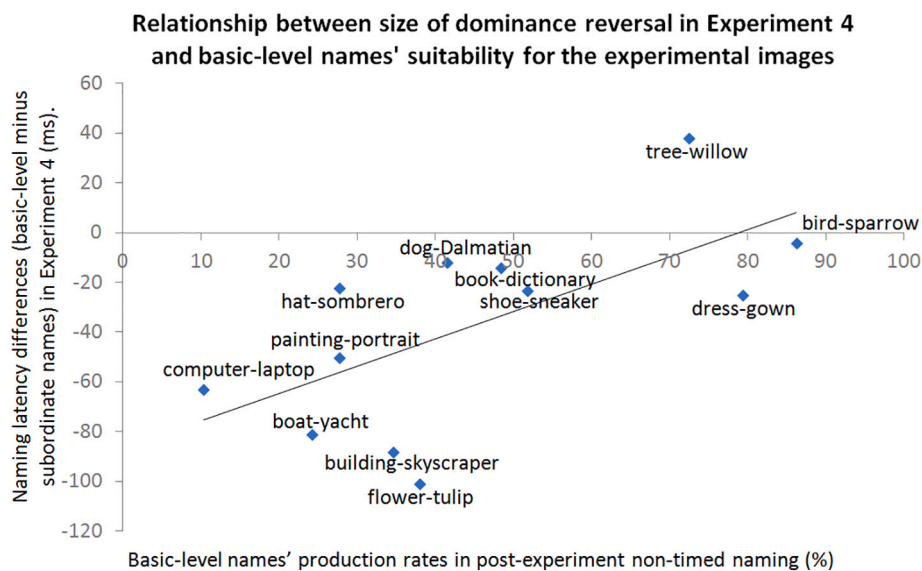


Fig. 13. Relationship between size of dominance reversal in Experiment 4 and basic-level names' suitability for the experimental images: the greater the suitability, the smaller the reversal.

Appendix C. Between-experiment comparisons

We compared the magnitudes of dominant deprioritization between monolinguals and nonbalanced bilinguals and between the two bilingual groups. The statistical models carrying out these comparisons had the same predictors as in the analyses of individual experiments (coded in the same way), with the following exceptions. The Language and Name Type predictors were collapsed into a single Name Type predictor (dominant, -0.5 , nondominant, 0.5). Also added were a Language group predictor and its interactions (Monolingual-bilingual comparison: nonbalanced bilinguals, coded as -0.39 , monolinguals, coded as 0.28 ; Bilingual comparison: nonbalanced bilinguals, coded as -0.45 , balanced bilinguals, coded as 0.55). Because of the partial difference in items between the bilingual and monolingual experiments, when comparing them we ran three separate models for each type of cost. The first model included all items (eight of which differed between the monolingual and the bilingual experiments). The second model included only the four shared items across the three experiments. The third model compared nonbalanced bilinguals with only one monolingual group (Experiment 4) and included only the six shared items between the two experiments. We pooled the monolingual data because there is no theoretical reason why the two monolingual groups should differ, but before doing so, we compared Experiments 3 and 4 directly to ensure pooling was warranted. This comparison indicated that switching, mixing and post-mixing costs did not differ between the two experiments.

The statistical indicator of whether a type of cost differed between the two groups is the three-way interaction between Trial or Block type, Name type and Language group. All effects and interactions of the Language group predictor are reported in Table 12 for the comparison of monolinguals with nonbalanced bilinguals, Table 13 for the comparison of nonbalanced with balanced bilinguals, and Table 14 for the comparison of the two monolingual experiments.

We further computed Bayes factors to directly assess the evidence for similarity of costs between the nonbalanced bilinguals and monolinguals, in those cases where we did not detect differences between them. We ran linear models (lmBF) with and without the critical three-way interaction term (Trial or Block type x Name type x Language group) for each of the three costs and in each of the three analyses over different item sets described above with the package BayesFactor in R (all models had random intercepts for subjects and items⁹). Bayes factors of more than 10 are interpreted as strong evidence for the theoretical prediction, while Bayes factors of less than 0.33 are interpreted as strong support for the null hypothesis (Jeffreys, 1961; Lee & Wagenmakers, 2013). The Bayes factors for the main analyses were $1329.04 \pm 23.96\%$ for mixing costs (where we did find a significant difference), $0.05 \pm 1.48\%$ for switching costs and $2.27 \pm 16.03\%$ for post-mixing costs. Across analyses of only shared item subsets, the largest value was 0.19. Taken together, these analyses do not present the strongest possible evidence that the deprioritization magnitudes do not differ between monolinguals and nonbalanced bilinguals. Where they did differ, the asymmetry was smaller for monolinguals than for nonbalanced bilinguals.

The monolingual data was not pooled for analyses of repetition benefits because repetition benefits did differ between Experiments 3 and 4 (there was a larger repetition benefit for subordinate than for basic-level names in all blocks in Experiment 4 but only marginally in the pre-mix block in Experiment 3; see Table 14). The comparison of nonbalanced bilinguals with monolinguals in Experiment 3 did not produce any significant interactions involving language group and repetition. Consistent with the patterns in the individual experiments, the comparison of nonbalanced bilinguals with monolinguals in Experiment 4 (with all items) showed that repetition benefits were more pronounced for the monolinguals than for the nonbalanced bilinguals.

Table 12

Comparison of nonbalanced bilinguals (Exp. 2) vs. monolinguals (Expts. 3 and 4) – effects and interactions involving the between-experiment predictor (Language group), and including all items.

Model	Predictors	Estimate	SE	t	p
Switching costs					
	Language group	-14.45	33.87	-.43	.67
	Switching trial type x Language group	-.34	10.45	-.03	.97
	Name type x Language group	-124.85	18.44	-6.77	< .001
	Switching trial type x Name x Language group	-1.23	20.71	-.06	.95
Mixing costs					
	Language group	-51.29	24.63	-2.08	.04
	Mixing trial type x Language group	69.59	25.58	2.72	.007
	Name type x Language group	-139.00	18.75	-7.42	< .001
	Trial type x Name x Language group	65.17	29.19	2.23	.03
Post-mixing costs on single naming					
	Language group	-59.35	23.92	-2.48	.01
	Single naming block type x Language group	56.58	22.61	2.50	.01
	Name type x Language group	-158.91	18.68	-8.51	< .001
	Single naming block type x Name x Language group	31.00	26.45	1.17	.24
Repetition benefits					
<i>Nonbalanced bilinguals (Experiment 2) vs. Monolinguals (Experiment 3)</i>					
	Language group	-53.06	21.29	-2.49	.01
	Block type x Language group	-23.86	17.00	-1.40	.16
	Name type x Language group	-100.63	13.63	-7.38	< .001
	Repetition x Language group	-2.01	3.54	-.57	.57
	Block type x Name type x Language group	-45.71	17.47	-2.62	.01
	Block type x Repetition x Language group	.06	6.28	.01	.99
	Name type x Repetition x Language group	-1.55	4.45	-.34	.73
	Block type x Name type x Repetition x Language group	7.72	8.53	.91	.37
<i>Nonbalanced bilinguals (Experiment 2) vs. Monolinguals (Experiment 4)</i>					
	Language group (Exp. 3 vs. Exp. 4)	1.47	23.17	.06	.95
	Block type x Language group	-36.23	16.13	-2.25	.03
	Name type x Language group	-90.85	13.56	-6.70	< .001
	Repetition x Language group	-5.72	3.67	-1.56	.12
	Block type x Name type x Language group	-37.56	17.42	-2.16	.03
	Block type x Repetition x Language group	.49	6.61	.07	.94
	Name type x Repetition x Language group	-10.29	4.86	-2.12	.04
	Block type x Name type x Repetition x Language group	7.15	9.62	.74	.46

⁹ The models we used for significance testing had larger random effects structures, but we consider it unlikely that the Bayes factor values we obtained critically depended on the models' random-effects structures.

Table 13

Comparison of nonbalanced bilinguals (Exp. 2) vs. balanced bilinguals (Exp. 1) – effects and interactions involving the between-experiment predictor (Language group).

Model	Predictors	Estimate	SE	t	p
Switching costs					
	Language group	4.32	26.16	.17	.87
	Switching trial type x Language group	20.04	7.54	2.66	.009
	Name type x Language group	-18.89	14.28	-1.32	.19
	Switching trial type x Name x Language group	-30.28	15.38	-1.97	.051
Mixing costs					
	Language group	-18.73	19.49	-.96	.34
	Mixing trial type x Language group	25.84	17.03	1.52	.13
	Name type x Language group	-39.62	14.82	-2.67	.009
	Trial type x Name x Language group	74.55	20.54	3.63	< .001
Post-mixing costs on single naming					
	Language group	-23.65	18.84	-1.26	.21
	Single naming block type x Language group	16.28	16.45	.99	.32
	Name type x Language group	-58.35	14.46	-4.04	< .001
	Single naming block type x Name x Language group	37.65	18.36	2.05	.04
Repetition benefits					
	Language group	-10.18	21.47	-.47	.64
	Block type x Language group	-29.93	14.75	-2.03	.04
	Name type x Language group	-34.13	12.79	-2.67	.009
	Repetition x Language group	-2.54	3.07	-.83	.41
	Block type x Name type x Language group	-42.67	15.45	-2.76	.006
	Block type x Repetition x Language group	.26	5.10	.05	.96
	Name type x Repetition x Language group	-1.96	4.30	-.46	.65
	Block type x Name type x Repetition x Language group	11.02	8.31	1.33	.19

Table 14

Comparison of monolinguals in Experiment 3 with monolinguals in Experiment 4 – effects and interactions involving the between-experiment predictor (Monolingual group).

<i>Monolinguals (Experiment 3 vs. Experiment 4)</i>					
Switching costs					
	Monolingual group	56.00	27.43	2.04	.04
	Switching trial type x Monolingual group	-3.44	7.03	-.49	.62
	Name type x Monolingual group	8.12	12.00	.68	.50
	Switching trial type x Name x Monolingual group	-7.16	15.41	-.47	.64
Mixing costs					
	Monolingual group	55.39	18.46	3.00	.003
	Mixing trial type x Monolingual group	4.85	22.51	.22	.83
	Name type x Monolingual group	15.94	11.93	1.34	.18
	Trial type x Name x Monolingual group	-16.76	22.73	-.74	.46
Post-mixing costs on single naming					
	Monolingual group	56.20	17.33	3.24	.002
	Single naming block type x Monolingual group	6.78	18.31	.37	.71
	Name type x Monolingual group	18.89	11.95	1.58	.12
	Single naming block type x Name x Monolingual group	-6.53	22.73	-.29	.77
Repetition benefits					
	Monolingual group	57.31	21.31	2.69	.008
	Block type x Monolingual group	-6.49	16.78	-.39	.70
	Name type x Monolingual group	12.73	10.60	1.20	.23
	Repetition x Monolingual group	-3.40	2.93	-1.16	.25
	Block type x Name type x Monolingual group	3.40	14.38	.24	.81
	Block type x Repetition x Monolingual group	2.32	5.46	.43	.67
	Name type x Repetition x Monolingual group	-8.97	4.32	-2.08	.04
	Block type x Name type x Repetition x Monolingual group	-3.37	8.29	-.41	.69

References

- Abutalebi, J., Annoni, J. M., Zimine, I., Pegna, A. J., Seghier, M. L., Lee-Jahnke, H., ... Khateb, A. (2008). Language control and lexical competition in bilinguals: An event-related fMRI study. *Cerebral Cortex*, 18, 1496–1505.
- Abutalebi, J., & Green, D. (2007). Bilingual language production: The neurocognition of language representation and control. *Journal of Neurolinguistics*, 20(3), 242–275.
- Baayen, R. H. (2008). *Analyzing linguistic data: A practical introduction to statistics using R*. Cambridge: Cambridge University Press.
- Barry, C., Hirsh, K. W., Johnston, R. A., & Williams, C. L. (2001). Age of acquisition, word frequency, and the locus of repetition priming of picture naming. *Journal of Memory and Language*, 44(3), 350–375.
- Baus, C., Costa, A., & Carreiras, M. (2013). On the effects of second language immersion on first language production. *Acta Psychologica*, 142(3), 402–409.
- Bialystok, E., Craik, F., & Luk, G. (2008). Cognitive control and lexical access in younger and older bilinguals. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 34(4), 859–873.
- Bialystok, E., Craik, F. I., & Freedman, M. (2007). Bilingualism as a protection against the onset of symptoms of dementia. *Neuropsychologia*, 45(2), 459–464.

- Bialystok, E., & Martin, M. M. (2004). Attention and inhibition in bilingual children: Evidence from the dimensional change card sort task. *Developmental Science*, 7(3), 325–339.
- Blanco-Elorrieta, E., & Pylkkänen, L. (2016). Bilingual language control in perception versus action: MEG reveals comprehension control mechanisms in anterior cingulate cortex and domain-general control of production in dorsolateral prefrontal cortex. *Journal of Neuroscience*, 36(2), 290–301.
- Bob, S. C., & Wodniecka, Z. (2013). Language switching in picture naming: What asymmetric switch costs (do not) tell us about inhibition in bilingual speech planning. *Journal of Cognitive Psychology*, 25(5), 568–585.
- Branzi, F. M., Della Rosa, P. A., Canini, M., Costa, A., & Abutalebi, J. (2016). Language control in bilinguals: Monitoring and response selection. *Cerebral Cortex*, 26(6), 2367–2380.
- Branzi, F. M., Martin, C. D., Abutalebi, J., & Costa, A. (2014). The after-effects of bilingual language production. *Neuropsychologia*, 52, 102–116.
- Brennan, S. E., & Clark, H. H. (1996). Conceptual pacts and lexical choice in conversation. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 22, 1482–1493.
- de Bruin, A., Roelofs, A., Dijkstra, T., & FitzPatrick, I. (2014). Domain-general inhibition areas of the brain are involved in language switching: fMRI evidence from trilingual speakers. *NeuroImage*, 90, 348–359.
- Brysbaert, M., & New, B. (2009). Moving beyond Kučera and Francis: A critical evaluation of current word frequency norms and the introduction of a new and improved word frequency measure for American English. *Behavior Research Methods*, 41, 977–990.
- Calabria, M., Costa, A., Green, D. W., & Abutalebi, J. (2018). Neural basis of bilingual language control. *Annals of the New York Academy of Sciences*, 1426(1), 221–235.
- Calabria, M., Hernández, M., Branzi, F. M., & Costa, A. (2012). Qualitative differences between bilingual language control and executive control: Evidence from task-switching. *Frontiers in Psychology*, 2, 399.
- Christoffels, I. K., Firk, C., & Schiller, N. O. (2007). Bilingual language control: An event-related brain potential study. *Brain Research*, 1147, 192–208.
- Christoffels, I. K., Ganushchak, L., & La Heij, W. (2016). When L1 suffers: Sustained, global slowing and the reversed language effect in mixed language context. In J. W. Schwieter (Ed.), *Cognitive Control and Consequences of Multilingualism* (pp. 171–192). Amsterdam/Philadelphia: John Benjamins.
- Colomé, A. (2001). Lexical activation in bilinguals' speech production: Language-specific or language-independent? *Journal of Memory and Language*, 45(4), 721–736.
- Costa, A., Caramazza, A., & Sebastián-Gallés, N. (2000). The cognate facilitation effect: Implications for models of lexical access. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 26, 1283–1296.
- Costa, A., Hernández, M., & Sebastián-Gallés, N. (2008). Bilingualism aids conflict resolution: Evidence from the ANT task. *Cognition*, 106, 59–86.
- Costa, A., & Santesteban, M. (2004). Lexical access in bilingual speech production: Evidence from language switching in highly proficient bilinguals and L2 learners. *Journal of Memory and Language*, 50(4), 491–511.
- Costa, A., Santesteban, M., & Ivanova, I. (2006). How do highly proficient bilinguals control their lexicalization process? Inhibitory and language-specific selection mechanisms are both functional. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 32(5), 1057–1074.
- Costa, A., Sebastián-Gallés, N., Miozzo, M., & Caramazza, A. (1999). The gender congruity effect: Evidence from Spanish and Catalan. *Language and Cognitive Processes*, 14(4), 381–391.
- Cuetos, F., Glez-Nosti, M., Barbon, A., & Brysbaert, M. (2011). SUBTLEX-ESP: frecuencias de las palabras españolas basadas en los subtítulos de las películas. *Psicológica*, 32(2), 133–144.
- Cutting, J. C., & Ferreira, V. S. (1999). Semantic and phonological information flow in the production lexicon. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 25(2), 318–344.
- Declerck, M., Koch, I., & Philipp, A. M. (2012). Digits vs. pictures: The influence of stimulus type on language switching. *Bilingualism: Language and Cognition*, 15(4), 896–904.
- Declerck, M., Grainger, J., Koch, I., & Philipp, A. M. (2017). Is language control just a form of executive control? Evidence for overlapping processes in language switching and task switching. *Journal of Memory and Language*, 95, 138–145.
- Declerck, M., Ivanova, I., Grainger, J., & Duñabeitia, J. A. (2020). Are similar control processes implemented during single and dual language production? Evidence from switching between speech registers and languages. *Bilingualism: Language and Cognition*, 23, 694–701.
- Declerck, M., Kleinman, D., & Gollan, T. H. (2020). Which bilinguals reverse language dominance and why? *Cognition*, 204, 104384.
- Declerck, M., & Philipp, A. M. (2015). A review of control processes and their locus in language switching. *Psychonomic Bulletin & Review*, 22(6), 1630–1645.
- Declerck, M., Philipp, A. M., & Koch, I. (2013). Bilingual control: Sequential memory in language switching. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 39(6), 1793–1806.
- Declerck, M., Stephan, D. N., Koch, I., & Philipp, A. M. (2015). The other modality: Auditory stimuli in language switching. *Journal of Cognitive Psychology*, 27(6), 685–691.
- Dylman, A. S., & Barry, C. (2018). When having two names facilitates lexical selection: Similar results in the picture-word task from translation distractors in bilinguals and synonym distractors in monolinguals. *Cognition*, 171, 151–171.
- Faul, F., Erdfelder, E., Buchner, A., & Lang, A.-G. (2009). Statistical power analyses using G*Power 3.1: Tests for correlation and regression analyses. *Behavior Research Methods*, 41, 1149–1160.
- Filippi, R., Karaminis, T., & Thomas, M. S. (2014). Language switching in bilingual production: Empirical data and computational modelling. *Bilingualism: Language and Cognition*, 17(2), 294–315.
- Fink, A., & Goldrick, M. (2015). Pervasive benefits of preparation in language switching. *Psychonomic Bulletin & Review*, 22(3), 808–814.
- Finkbeiner, M., Almeida, J., Janssen, N., & Caramazza, A. (2006). Lexical selection in bilingual speech production does not involve language suppression. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 32(5), 1075–1089.
- Finkbeiner, M., Gollan, T. H., & Caramazza, A. (2006). Lexical access in bilingual speakers: What's the (hard) problem? *Bilingualism: Language and Cognition*, 9(2), 153–166.
- Forster, K. I., & Forster, J. C. (2003). DMDX: A windows display program with millisecond accuracy. *Behavior Research Methods, Instruments, & Computers*, 35(1), 116–124.
- Francis, W. S. (2005). Bilingual semantic and conceptual representation. In J. F. Kroll, & A. M. B. de Groot (Eds.), *Handbook of bilingualism: Psycholinguistic approaches* (pp. 251–267). New York: Oxford University Press.
- Gollan, T. H., & Ferreira, V. S. (2009). Should I stay or should I switch? A cost-benefit analysis of voluntary language switching in young and aging bilinguals. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 35(3), 640–665.
- Gollan, T. H., & Goldrick, M. (2016). Grammatical constraints on language switching: Language control is not just executive control. *Journal of Memory and Language*, 90, 177–199.
- Gollan, T. H., Montoya, R. I., Fennema-Notestine, C., & Morris, S. K. (2005). Bilingualism affects picture naming but not picture classification. *Memory & Cognition*, 33(7), 1220–1234.
- Gollan, T. H., Sandoval, T., & Salmon, D. P. (2011). Cross-language intrusion errors in aging bilinguals reveal the link between executive control and language selection. *Psychological Science*, 22(9), 1155–1164.
- Gollan, T. H., Kleinman, D., & Wierenga, C. E. (2014). What's easier: Doing what you want, or being told what to do? Cued versus voluntary language and task switching. *Journal of Experimental Psychology: General*, 143(6), 2167.
- Gollan, T. H., Schotter, E. R., Gomez, J., Murillo, M., & Rayner, K. (2014). Multiple levels of bilingual language control: Evidence from language intrusions in reading aloud. *Psychological Science*, 25(2), 585–595.
- Gollan, T. H., Weissberger, G. H., Runnqvist, E., Montoya, R. I., & Cera, C. M. (2012). Self-ratings of spoken language dominance: A multilingual naming test (MINT) and preliminary norms for young and aging Spanish–English bilinguals. *Bilingualism: Language and Cognition*, 15(3), 594–615.
- Green, D. W. (1998). Mental control of the bilingual lexico-semantic system. *Bilingualism: Language and Cognition*, 1(2), 67–81.
- Griffin, Z. M., & Bock, K. (1998). Constraint, word frequency, and the relationship between lexical processing levels in spoken word production. *Journal of Memory and Language*, 38(3), 313–338.
- Grosjean, F. (2001). The bilingual's language modes. In J. L. Nicol (Ed.), *Explaining linguistics. One mind, two languages: Bilingual language processing* (pp. 1–22). Oxford: Blackwell Publishing.
- Guo, T., Liu, H., Misra, M., & Kroll, J. F. (2011). Local and global inhibition in bilingual word production: fMRI evidence from Chinese–English bilinguals. *NeuroImage*, 56(4), 2300–2309.
- Hantsch, A., Jescheniak, J. D., & Schriefers, H. (2005). Semantic competition between hierarchically related words during speech planning. *Memory & Cognition*, 33, 984–1000.
- Hanulová, J., Davidson, D. J., & Indefrey, P. (2011). Where does the delay in L2 picture naming come from? Psycholinguistic and neurocognitive evidence on second language word production. *Language and Cognitive Processes*, 26(7), 902–934.
- Hoshino, N., & Kroll, J. F. (2008). Cognate effects in picture naming: Does cross-language activation survive a change of script? *Cognition*, 106(1), 501–511.
- Howard, D., Nickels, L., Coltheart, M., & Cole-Virtue, J. (2006). Cumulative semantic inhibition in picture naming: Experimental and computational studies. *Cognition*, 100(3), 464–482.
- Jaeger, T. F. (2008). Categorical data analysis: Away from ANOVAs (transformation or not) and towards logit mixed models. *Journal of Memory and Language*, 59(4), 434–446.
- Jeffreys, H. (1961). *The theory of probability*. New York, NY: Oxford University Press.
- Jin, Z. L., Zhang, J. X., & Li, L. (2014). Endogenous language control in Chinese-English switching: An event-related potentials study. *Neuroscience Bulletin*, 30(3), 461–468.
- Kirk, N. W., Kempe, V., Scott-Brown, K. C., Philipp, A., & Declerck, M. (2018). Can monolinguals be like bilinguals? Evidence from dialect switching. *Cognition*, 170, 164–178.
- Kleinman, D., & Gollan, T. H. (2016). Speaking two languages for the price of one: Bypassing language control mechanisms via accessibility-driven switches. *Psychological Science*, 27(5), 700–714.
- Kleinman, D., & Gollan, T. H. (2018). Inhibition accumulates over time at multiple processing levels in bilingual language control. *Cognition*, 173, 115–132.
- Kousaie, S., & Phillips, N. A. (2012). Ageing and bilingualism: Absence of a “bilingual advantage” in Stroop interference in a nonimmigrant sample. *Quarterly Journal of Experimental Psychology*, 65(2), 356–369.
- Kroll, J. F., & Stewart, E. (1994). Category interference in translation and picture naming: Evidence for asymmetric connections between bilingual memory representations. *Journal of Memory and Language*, 33(2), 149–174.
- Kuipers, J. R., La Heij, W., & Costa, A. (2006). A further look at semantic context effects in language production: The role of response congruency. *Language and Cognitive Processes*, 21(7–8), 892–919.

- La Heij, W. (2005). Selection processes in monolingual and bilingual lexical access. In J. F. Kroll, & A. M. B. de Groot (Eds.), *Handbook of bilingualism: Psycholinguistic approaches* (pp. 289–307). New York: Oxford University Press.
- Lee, M. D., & Wagenmakers, E. J. (2013). *Bayesian data analysis for cognitive science: A practical course*. New York, NY: Cambridge University Press.
- Levett, W. J., Roelofs, A., & Meyer, A. S. (1999). A theory of lexical access in speech production. *Behavioral and Brain Sciences*, 22(1), 1–38.
- Li, C., & Gollan, T. H. (2018). Cognates facilitate switches and then confusion: Contrasting effects of cascade versus feedback on language selection. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 44(6), 974.
- Linck, J. A., Kroll, J. F., & Sunderman, G. (2009). Losing access to the native language while immersed in a second language: Evidence for the role of inhibition in second-language learning. *Psychological Science*, 20(12), 1507–1515.
- Macizo, P., Bajo, T., & Paolieri, D. (2012). Language switching and language competition. *Second Language Research*, 28(2), 131–149.
- Mahon, B. Z., Costa, A., Peterson, R., Vargas, K. A., & Caramazza, A. (2007). Lexical selection is not by competition: A reinterpretation of semantic interference and facilitation effects in the picture-word interference paradigm. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 33(3), 503–535.
- Melinger, A. (2018). Distinguishing languages from dialects: A litmus test using the picture-word interference task. *Cognition*, 172, 73–88.
- Melinger, A. (2021). Do elevators compete with lifts? Selecting dialect alternatives. *Cognition*, 206, 104471.
- Meuter, R. F., & Allport, A. (1999). Bilingual language switching in naming: Asymmetrical costs of language selection. *Journal of Memory and Language*, 40(1), 25–40.
- Misra, M., Guo, T., Bobb, S. C., & Kroll, J. F. (2012). When bilinguals choose a single word to speak: Electrophysiological evidence for inhibition of the native language. *Journal of Memory and Language*, 67(1), 224–237.
- Mosca, M., & Clahsen, H. (2016). Examining language switching in bilinguals: The role of preparation time. *Bilingualism: Language and Cognition*, 19(2), 415–424.
- Norman, D. A., & Shallice, T. (1986). Attention to action: Willed and automatic control of behavior. In R. J. Davidson, G. E. Schwartz, & D. Shapiro (Eds.), *Vol. 4. Consciousness and self-regulation: Advances in research and theory* (pp. 1–18). New York: Plenum.
- Nozari, N., & Hepner, C. R. (2019). To select or to wait? The importance of criterion setting in debates of competitive lexical selection. *Cognitive Neuropsychology*, 36(5–6), 193–207.
- Oppenheim, G. M., Dell, G. S., & Schwartz, M. F. (2010). The dark side of incremental learning: A model of cumulative semantic interference during lexical access in speech production. *Cognition*, 114(2), 227–252.
- Paap, K. R., & Greenberg, Z. I. (2013). There is no coherent evidence for a bilingual advantage in executive processing. *Cognitive Psychology*, 66(2), 232–258.
- Peeters, D., & Dijkstra, T. (2018). Sustained inhibition of the native language in bilingual language production: A virtual reality approach. *Bilingualism: Language and Cognition*, 21(5), 1035–1061.
- Peeters, D., Runnqvist, E., Bertrand, D., & Grainger, J. (2014). Asymmetrical switch costs in bilingual language production induced by reading words. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 40(1), 284–292.
- Philipp, A. M., Gade, M., & Koch, I. (2007). Inhibitory processes in language switching: Evidence from switching language-defined response sets. *European Journal of Cognitive Psychology*, 19(3), 395–416.
- Poullisse, N. (1999). *Slips of the tongue: Speech errors in first and second language production*. Amsterdam: Benjamins.
- Poullisse, N., & Bongaerts, T. (1994). First language use in second language production. *Applied Linguistics*, 15(1), 36–57.
- Prior, A., & Gollan, T. H. (2011). Good language-switchers are good task-switchers: Evidence from Spanish–English and Mandarin–English bilinguals. *Journal of the International Neuropsychological Society*, 17(4), 682–691.
- Roelofs, A. (1992). A spreading-activation theory of lemma retrieval in speaking. *Cognition*, 42(1–3), 107–142.
- Roelofs, A. (2003). Goal-referenced selection of verbal action: Modeling attentional control in the Stroop task. *Psychological Review*, 110(1), 88–125.
- Runnqvist, E., Strijkers, K., Alario, F. X., & Costa, A. (2012). Cumulative semantic interference is blind to language: Implications for models of bilingual speech production. *Journal of Memory and Language*, 66(4), 850–869.
- Runnqvist, E., Strijkers, K., & Costa, A. (2019). Error-based learning and lexical competition in word production: Evidence from multilingual naming. *PLoS One*, 14(3), Article e0213765.
- Runnqvist, E., Strijkers, K., Sadat, J., & Costa, A. (2011). On the temporal and functional origin of L2 disadvantages in speech production: A critical review. *Frontiers in Psychology*, 2, 379.
- Seo, R., Stocco, A., & Prat, C. S. (2018). The bilingual language network: Differential involvement of anterior cingulate, basal ganglia and prefrontal cortex in preparation, monitoring, and execution. *NeuroImage*, 174, 44–56.
- Starreveld, P. A., & La Heij, W. (1995). Semantic interference, orthographic facilitation, and their interaction in naming tasks. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 21(3), 686–698.
- Strijkers, K., Costa, A., & Thierry, G. (2010). Tracking lexical access in speech production: Electrophysiological correlates of word frequency and cognate effects. *Cerebral Cortex*, 20(4), 912–928.
- Van Assche, E., Duyck, W., & Gollan, T. H. (2013). Whole-language and item-specific control in bilingual language production. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 39(6), 1781–1792.
- Verhoef, K., Roelofs, A., & Chwilla, D. J. (2009). Role of inhibition in language switching: Evidence from event-related brain potentials in overt picture naming. *Cognition*, 110(1), 84–99.
- Verhoef, K. M., Roelofs, A., & Chwilla, D. J. (2010). Electrophysiological evidence for endogenous control of attention in switching between languages in overt picture naming. *Journal of Cognitive Neuroscience*, 22(8), 1832–1843.
- Von Bastian, C. C., Souza, A. S., & Gade, M. (2016). No evidence for bilingual cognitive advantages: A test of four hypotheses. *Journal of Experimental Psychology: General*, 145(2), 246–258.
- Wang, Y., Kuhl, P. K., Chen, C., & Dong, Q. (2009). Sustained and transient language control in the bilingual brain. *NeuroImage*, 47(1), 414–422.
- Weiss, S. (2018, October). 16. The Cut: The embarrassing private languages of couples. Retrieved from <https://www.thecut.com/2018/10/its-complicated-embarrassing-couples-languages.html>.
- Wheeldon, L. R., & Monsell, S. (1994). Inhibition of spoken word production by priming a semantic competitor. *Journal of Memory and Language*, 33(3), 332–356.